Farm Household Misallocation

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Abstract

Agricultural markets often fail to allocate resources efficiently across farm households in developing countries. To address these, policymakers require knowledge of which markets fail and how the distortions they generate are correlated. Using data from rural Thailand, I characterize how distortions in individual input (e.g. land, labor, equipment) and financial (credit and insurance) markets each contribute to misallocation. I use moments in household consumption and production data to separately identify these distortions and develop a novel method using them to structurally estimate the production function. I find that the efficient allocation would increase aggregate productivity by 11.3% relative to the status quo, while only 5.9% (4.1%) gains could be achieved by eliminating input (financial) distortions in isolation. Positive interaction effects from addressing multiple distortions simultaneously account for the remaining 1.3% TFP gains. There also appear to be sharply decreasing marginal returns to addressing a single distortion in isolation, suggesting that policies overcoming inefficiencies in a broad set of markets would outperform attempts to eliminate a single one. Meanwhile, other common methods would produce larger estimates of misallocation and suggest that a financial market intervention would decrease aggregate productivity. Accounting for multiple correlated distortions is therefore crucial for measuring misallocation and designing policies to address it.

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1 Introduction

Agricultural productivity is up to sixty times higher in the richest than in the poorest countries (Gollin et al., 2014). A leading explanation for these gaps is that local markets in low income countries fail to allocate resources efficiently across producers (Adamopoulos & Restuccia, 2014). This cross-country evidence is qualitatively consistent with extensive micro evidence of pervasive failures in both factor (e.g. land, labor, fertilizer) and financial (credit, insurance) markets, often operating simultaneously.[1] Yet effective policy requires not just quantifying the magnitudes of individual distortions arising from these market failures, but also identifying how they amplify or offset one another in equilibrium. Following the theory of the second best, the effects of reducing a single distortion are generally ambiguous—and can even reduce productivity—depending on the joint distribution of all distortions (Lipsey & Lancaster, 1956; Baqaee & Farhi, 2020).

This paper develops a novel approach to identifying and estimating this joint distribution, which I use to diagnose the sources of misallocation in Thai agriculture. However, this requires overcoming a fundamental identification problem: these distortions cannot be separately identified from production data alone (Hsieh & Klenow, 2009). Intuitively, data on inputs and outputs cannot reveal whether a household uses less of an input because it faces a higher price for it or cannot access credit or insurance. Moreover, dispersion in measured productivity could reflect measurement error rather than true distortions—a problem shown to severely inflate estimates of misallocation in agricultural settings (Gollin & Udry, 2021). The key insight that enables identification is that unlike typical firms, farm households are also consumers. Under imperfect markets, household consumption enters investment decisions and therefore contains information about how production is distorted (Benjamin, 1992). I leverage this fact to identify how credit constraints and uninsured risk distort pro-

[1] See Magruder (2018) and Suri and Udry (2022) for recent overviews. Bold et al. (2017), Breza et al. (2021), Caunedo and Kala (2021), Diop (2023), Goldstein and Udry (2008), Karlan et al. (2014), and Mobarak and Rosenzweig (2013) provide excellent examples of each of these market failures, respectively. Emerick et al. (2016) and Jones et al. (2022) are examples providing experimental evidence on how these market failures can compound each other.

ductive choices distinctly from frictions in input markets. In particular, credit constraints are reflected in systematically lower consumption at planting than at harvest (reflecting the inability to smooth consumption across time by borrowing against future harvests). Meanwhile, uninsured production risk causes consumption at harvest to depend on the realization of production shocks. On the other hand, input frictions function like a tax or subsidy and can be identified from dispersion in input *composition* across similar households. This enables me to estimate these distortions from general first order conditions for the household's value function under non-separability and characterize their joint distribution.

I jointly estimate the production function and these distortions using 196 months of panel data on inputs, output and consumption from Thai farm-households in two steps. In the first step, I leverage two sets of moment equations from a general farm-household model. The first moment condition is simply the physical production function, relating *realized* output to inputs. The second is the demand function for each input, which under perfect markets would imply that households equate the *expected* marginal revenue product of each input to a common factor price.[2] These are essentially asset pricing equations, in which the return structure of the assets (inputs) is governed by the physical production function—and weighted by a stochastic discount factor (SDF) under financial frictions. The difference between these SDF-adjusted returns and the market price for each input identifies the idiosyncratic friction a household faces, which can be recovered using a canonical GMM application (Hansen & Singleton, 1982).

The second step isolates actual (ex ante) distortions from the realization of ex-post shocks to productivity and consumption, the latter of which cannot be considered misallocation (Gollin & Udry, 2021). This is a difficult problem, as it requires taking a stand on the joint distribution of production shocks and post-harvest consumption from the household's perspective. I take a data-driven approach to this problem, instead of placing additional para-

[2]The simplest example of this is calibrating Cobb-Douglas coefficients to observed revenue shares. However, these are not valid under imperfect markets because firms do not maximize expected profits and do not face common prices.

metric structure on households' expectations—a complicated function of potentially many unobserved state variables. I do so by predicting realized productivity and consumption from the rich set of features from households' ex-ante information sets available in the Thai data—essentially trying to approximate the prediction problem the household faces. This helps me isolate the wedge due to financial frictions along with anticipated TFP from ex-post shocks—something which any analysis of misallocation must make assumptions about. As I discuss in Section 4, this may produce somewhat conservative estimates of financial misallocation. I therefore present alternative results using "naive" approaches that bypass this prediction step entirely—reassuringly, the estimates are similar, suggesting the attenuation problem is not too serious empirically.

I then use estimates of the production function and distortions to calculate aggregate TFP under the observed allocation, and decompose misallocation into that due to factor market frictions, financial market frictions, and their interaction. An important advantage of this decomposition, which crucially is only feasible when both input and financial distortions are well-specified, is that it diagnoses the sources of misallocation while remaining agnostic towards the specific institutions that generate distortions. These distortions have many potential, possibly simultaneous, causes and conclusions may depend on which ones a model specifies.[3] Instead, my method allows me to diagnose how distortions in each market affect aggregate productivity without strong assumptions about their root causes.

I estimate overall misallocation — which I define as the TFP increase from the observed to efficient allocation — of 11.3%. This is lower than estimates of total misallocation of 19% in Shenoy (2017) from Thailand (albeit using different methodologies and data), and substantially lower than estimates of 53% from China (Adamopoulos et al., 2022), 97% from Ethiopia (Chen et al., 2022), 259% from Malawi (Chen et al., 2023), and 286% from

[3]For example, recent empirical work has identified expropriation risk Goldstein and Udry (2008), incomplete contracting Burchardi et al. (2019), an explicit cap on landholdings Adamopoulos and Restuccia (2020), lack of titling Chen et al. (2022), land fragmentation Bryan et al. (2022), and others, as contributing to imperfect land markets. It would be impossible to capture all of these explicitly in a single model.

Uganda (Aragon et al., 2022).[4] These differences are due to both setting and methodology—common methods in the literature would produce more than twice as large estimates of total misallocation (28.8%). I show that this is because estimating the full set of distortions allows me to avoid inferring some from a residual which contains both true misallocation and measurement error in inputs, which appears as a spurious distortion.

I then decompose these potential TFP gains into the effects of eliminating either set of distortions in isolation and the interaction effect from eliminating them simultaneously.[5, 6] I find that removing financial distortions while holding observed input wedges fixed would achieve 4.1% TFP gains relative to the observed allocation while removing input distortions alone would achieve 5.9% gains. These sum to less than the 11.3% gains from eliminating both sets of distortions together, implying a positive interaction effect of 1.3%. A less conservative approach to measuring financial misallocation yields 14.7% total gains, of which 7.3% comes from financial misallocation alone, while relaxing input frictions alone yields 4.5%. In either case, the interaction effect is positive because, empirically more financially constrained households are effectively subsidized in input markets.[7] However, this modest interaction effect masks sharply diminishing marginal returns to (uniformly) reducing distortions in a single market in isolation. Together these findings suggest that a multi-pronged policy approach addressing multiple frictions together could achieve larger gains from the same amount of resources than focusing intensively on a single market failure. Meanwhile, other methods would incorrectly suggest that removing financial frictions lowers aggregate productivity. While my approach requires additional modeling assumptions, this comparison

[4] This is more striking, given that the latter three papers only consider land misallocation.

[5] While the method identifies distortions for each input as well as credit and risk, I present most of the main results grouping them into two broad classes—input and financial distortions—to keep the exposition compact.

[6] Note that this decomposition does not take into account the possible effects of relaxing one distortion on another — for example reducing land frictions through tenure reform could reduce credit constraints if it allows households to use land as collateral (Manysheva, 2021). However, modeling these potential effects would require taking strong stances on the types of institutions responsible for distortions and is beyond the scope of this paper.

[7] This is consistent with evidence that poorer households over-supply labor to their own farms because the shadow value of their time is lower Dillon et al. (2019) and Jones et al. (2022).

illustrates how leaving distortions unmodeled can generate not just inflated but qualitatively misleading policy implications.

This paper's main contribution is developing a framework and estimation strategy to attribute misallocation to failures in distinct markets. Recent advances in the misallocation literature (e.g. Carrillo et al., 2023; Sraer & Thesmar, 2023; Hughes & Majerovitz, 2023) show how misallocation can nonparametrically be estimated from (quasi-)experimental variation but are generally unable to trace misallocation to its different sources. There is also a growing literature applying quantitative misallocation models to microdata in agriculture (Adamopoulos & Restuccia, 2020; Adamopoulos et al., 2022, 2024; Aragon et al., 2022; Chari et al., 2021; Chen et al., 2017, 2022, 2023; Donovan, 2021; Gottlieb & Grobovšek, 2019; Manysheva, 2021; Shenoy, 2017). However, these papers typically model a single distortion in isolation or combine all distortions into a composite wedge. Notable exceptions are Manysheva (2021), who models the explicit dependence of credit constraints and land distortions through the collateral channel, and Shenoy (2017) who, also in Thailand, derives bounds for input and financial misallocation under assumptions on the joint distribution of distortions.[8] In contrast, I estimate a more complete range of distortions that captures how they may compound or offset each other, which has important implications for policy.

Second, I contribute to the recent literature on how measurement error can inflate estimates of misallocation by using consumption alongside production data to separate between financial frictions and input mismeasurment. Rotemberg and White (2021) and Bils et al. (2021) find large upward biases due to measurement error in U.S. and Indian manufacturing. Meanwhile, Gollin and Udry (2021) argue that up to 70% of observed productivity dispersion in Ugandan and Tanzanian agriculture is due to measurement error and unobserved heterogeneity. This is supported by evidence of large and systematic measurement error in survey measures of agricultural land, labor, and output (Abay et al., 2019, 2021; Arthi et al., 2018; Desiere & Jolliffe, 2018).

[8] Outside of agriculture, David and Venkateswaran (2019) attempt to disentangle frictions such as imperfect information and adjustment costs among Chinese manufacturing firms.

Third, I contribute to the literature on production function estimation when input choices are distorted. This has been done in previous work to address adjustment costs (Asker et al., 2014), input price dispersion (De Loecker et al., 2016; Grieco et al., 2016), and markups (Asker et al., 2019; Cairncross et al., 2023; De Loecker & Warzynski, 2012), but not the types of distortions that farm households are likely to face, such as uninsured risk. My approach is to use a theory-consistent model of households' constrained optimal behavior to identify the production function given how input and financial frictions enter first-order conditions. Doing so ensures unobserved shocks' effects on input demands are subsumed by households' potentially distorted choices of consumption and investment.

The rest of the paper is organized as follows: Section 2 provides more information about the Thai data and context. In Section 3, I present the theoretical framework and derive expressions for financial and input wedges at the household level, showing how they map to aggregate misallocation. Section 4 presents the estimation framework I develop and the results. Section 5 describes the decomposition of misallocation and Section 6 concludes.

2 Empirical Setting and Data

I use the 196 rounds of the Townsend Thai Monthly Survey from 1998 to 2014. Thailand during the sample period— covering the aftermath of the Asian financial crisis and its ascension to upper-middle-income status in 2011— offers a useful benchmark for understanding misallocation in developing-country agriculture. Unlike settings where the most dramatic estimates of agricultural misallocation have been obtained, Thailand lacks the severe institutional distortions such as explicit caps on landholdings (e.g. Adamopoulos & Restuccia, 2020), de jure expropriation (Adamopoulos et al., 2024) or financial autarky in other settings. There is also substantial regional heterogeneity in the sample: The four provinces in the sample span Thailand's Central and Northeastern regions, which differ substantially in agricultural commercialization and market integration. The Central region (Chachoengsao

and Lopburi) is closer to Bangkok, more commercialized, and better connected to output and input markets. The Northeastern region (Buriram and Sisaket) is poorer, more remote, and characterized by thinner markets and greater reliance on subsistence production. Nevertheless, Thai agriculture remains characterized throughout by rural smallholders rather than industrial farms. Therefore, Thai levels of misallocation may serve as a more reasonable benchmark for less developed countries than perfect markets or US agriculture.

The Townsend Thai Data is a 196-month panel dataset covering approximately 791 households in 16 villages across 4 townships, each in a different province. Each *month* the survey collects detailed information on agricultural production, household consumption, labor supply, and financial transactions, making it uniquely suited for studying input and financial misallocation. 621 of the 791 households engage in agriculture during the sample period. Over 68% of plots are cultivated with rice. In addition to crop production, households earn income from wages, livestock and aquaculture, and other businesses—the average agricultural household earns slightly less than half its income from crop cultivation.[9]

For the main analysis, I treat the township as the level of aggregation, since villages within townships are often quite integrated Kaboski and Townsend (2011) and Samphantharak and Townsend (2018). I focus on the sample of households cultivating rice during the main season. I drop all plots that do not report using land or labor as well as those that take more than 8 months between planting and harvest, although I also show robustness to using all annual crops during the main season.

2.1 Agricultural production and inputs

Throughout the main analysis, I consider seven inputs: land, fertilizer, equipment (mainly tractors, power tillers, and pumps), seed, and labor at three stages of production (planting, weeding, and harvesting).[10] The data contain quantity and expenditure/sale value (or the

[9]While the model in Section 3 is a one-sector model, the estimation procedure I develop explicitly accounts for the fact that households obtain returns from other sources that may be correlated with those of crop cultivation. In fact, this is essential for matching the data.

[10]Together these inputs make up over 98% of the total market value of inputs used.

household's estimate of market value where an input is not transacted) and quantity data on each of these inputs and output at a monthly frequency, which allows me to differentiate labor by production stage. This helps capture some of the seasonality in rural labor markets—tightness during planting and harvesting but slack at other times—without imposing a fully sequential production function. I aggregate inputs to the farm-year level, since the model implies that shadow prices of inputs and consumption apply to all plots cultivated by a household at a given time. This yields a panel of 6,223 farm-season observations across 16 years.

Table C3 presents summary statistics for agricultural production. Markets for all seven inputs exist, but vary considerably in depth. Land and labor markets are much more active in the Central region and appear quite thin in the Northeast. About 16% of farms rent land in a given season, but this masks substantial regional heterogeneity: nearly 40% of farms rent land in Chachoengsao while only 2.5% rent land in Sisaket. The average farm hires about 28% of its labor input, although more than two-thirds of farms hire some labor in a given season. Fertilizer, commercial seed, and mechanization are widespread and frequently acquired from outside the township.

Table C4 presents coefficients of variation for input and output prices within the average township-year. There is very little price variation for fertilizer, seed, and rice, but substantial dispersion in wages, land rents, and equipment rental rates. This pattern supports my assumption that output is perfectly tradable within townships—so that the law of one price holds—while land, labor, and equipment markets may be subject to household-specific frictions or rations.[11] I address concerns that some of this variation could be due to differential input quality in Section 4 by showing sensitivity to using hedonic approaches to price different attributes of plots, machines and workers.

[11] I also allow for frictions in fertilizer and seed markets, but consistent with variation in observed prices, find that these are much smaller than those in land, labor and equipment markets.

2.2 Consumption and Financial Markets

To estimate financial wedges, I require measures of the marginal utility of consumption at planting and harvest. The survey records monthly expenditures on 47 food and non-food goods, from which I estimate household-month marginal utilities using the procedure described in Section 4.1.2. Households actively participate in both formal and informal finance, obtaining loans from government banks, credit schemes, neighbors, and informal moneylenders. However, most loans are small and only 5.7% are collateralized, suggesting that credit markets are far from complete. This is consistent with evidence of credit constraints using the Townsend Data Kaboski and Townsend (2011), Kaboski and Townsend (2012), and Karaivanov and Townsend (2014). There is also widespread exchange of gifts among households, which several papers suggest form imperfect but relatively effective risk-sharing networks (Kinnan & Townsend, 2012; Samphantharak & Townsend, 2018; Kinnan et al., 2024).

2.3 Suggestive Evidence of Imperfect Markets in Thailand

Before imposing the structure of my model, I implement two canonical tests of complete markets. First, Townsend (1994) provides a test of full insurance: under complete risk-sharing, a regression of log consumption on log income with household and village-year fixed effects should yield a coefficient of zero, since idiosyncratic income shocks should be fully absorbed by the village risk-sharing network.[12] Second, Benjamin (1992) tests the joint null of complete markets by examining whether household composition—which affects preferences but not technology—predicts labor use. Under complete markets, production and consumption decisions are separable, so household demographics should be uncorrelated with input choices. Rejection is typically suggestive of labor market frictions causing households with larger labor endowments to supply more labor to their own farms, although it can in theory be associated with any market failure.

[12] Ligon (2025) illustrates the importance of using a measure of the marginal utility of consumption instead of log consumption if preferences are non-homothetic and shocks are covariate.

Table C1 presents the results. Column (1) reports the Townsend test: the coefficient on log income is 0.056 — in-line with recent estimates from India (Meghir et al., 2022) and highly significant. Columns (2) and (3) report the Benjamin test using household size and detailed age-sex composition, respectively; both reject separability, the latter at the 5% level.

While these regression coefficients do not have structural interpretations, their correlation can be informative about how frictions may interact. I run both tests separately for 64 village $\times$ 4-year blocks and plot the coefficients against each other in Figure C1. If financial and input market frictions were independent across villages, we would expect no systematic relationship. Instead, the coefficients are negatively correlated: villages where consumption is more sensitive to income shocks (indicating worse insurance) also exhibit less dependence of labor use on household composition (indicating more distorted labor markets).

This correlation is consistent with financial and input frictions partially offsetting each other— households that cannot smooth consumption may also be unable to hire labor at market wages, forcing them to rely on family labor regardless of its opportunity cost. However, these reduced-form tests cannot quantify the contribution of each friction to aggregate misallocation or determine whether the frictions offset or reinforce each other in equilibrium. Doing so requires the structural approach I develop in the following sections.

3 Model

I apply the workhorse farm-household model (Singh et al., 1986) to characterize how frictions in financial and input markets generate distinct wedges in households' input demands. I then show how these household-level wedges combine in equilibrium to distort the allocation away from the efficient one a social planner would choose. The model yields general moment conditions that permit the identification and estimation of distortions from data on production and consumption.

The goal is to measure misallocation and diagnose which markets contribute to it, while

remaining agnostic about the specific institutions that generate distortions. This agnosticism toward root causes is deliberate: distortions in any given market may arise from many sources simultaneously. For example, financial frictions could stem from limited commitment in informal insurance networks (Ligon et al., 2002), moral hazard in credit markets (Karlan & Zinman, 2009), or simply the absence of formal financial institutions (Karaivanov & Townsend, 2014). Likewise, land market distortions could reflect expropriation risk (Goldstein & Udry, 2008), incomplete contracting (Burchardi et al., 2019), explicit caps on holdings (Adamopoulos & Restuccia, 2020), or lack of titling (Chen et al., 2022). My approach captures the aggregate effect of multiple frictions on input allocation without requiring strong assumptions on which institutions generate them. The cost of this generality is that I cannot model how relaxing one distortion might affect another—for instance, how land titling might ease credit constraints through the collateral channel (Manysheva, 2021). I return to this limitation when interpreting the decomposition results in Section 5.

3.1 Setup

The environment is standard. Households i have time-separable preferences over a vector of consumption goods (c) and leisure (l), with per-period utility $u(c_{it}, l_{it})$ and maximize expected lifetime utility, which I show in Appendix B. They produce a single homogeneous agricultural good, and may hold a portfolio of assets with (possibly) stochastic returns.[13] I assume production uses Cobb-Douglas technology with decreasing returns to scale $\gamma \equiv \sum_k \alpha_k < 1$:[14]

$$Y_{it+1} = A_{it}\varphi_{it+1} \prod_k Q_{ikt}^{\alpha_k} \tag{1}$$

in which Q_{it} is a vector of K inputs applied by i at time t, A_{it} is anticipated productivity

[13] This is a flexible way of capturing the fact that households have other incomes sources beyond agriculture without having to explicitly model these sectors.

[14] Since there are no fixed factors, constant returns to scale would imply the efficient allocation is degenerate, with only the produce with the highest A_{it} producing. However, I do not impose the constraint $\gamma < 1$ during the estimation.)

(known when inputs are chosen)[15] and φ_{it+1} is an unanticipated mean 1 shock realized at time $t+1$, prior to harvesting output Y_{it+1}.[16]

Each period, households choose consumption, input quantities for the following period's harvest, and asset holdings, subject to a budget constraint. They may also face binding credit constraints that prevent borrowing against future output, incomplete insurance that leaves consumption exposed to production shocks, and frictions in input markets that cause effective prices to deviate from market prices. The full dynamic program is in Appendix A.

A first set of FOCs requires that households set the marginal utility of consumption from each good j equal to its price p_{jt} times the Lagrange multiplier associated with the budget constraint, λ_{it}, which can be thought of as the marginal utility of expenditure.[17]

$$u_j(c_{it}, l_{it}) = p_{jt}\lambda_{it} \tag{2}$$

Note that the budget constraint must hold in each state of the world, and thus λ_{it} is a random variable as of periods before t (absent full insurance). Estimates of λ_{it} will be key for estimating financial distortions in what follows.

A second set of FOCs states that households set the expected marginal product of each input weighted by the stochastic discount factor associated with each realization of production shocks to its shadow price. Under Cobb-Douglas, this takes the form

$$q_{ikt}w_{ikt} = \delta\alpha_k \frac{\mathbb{E}_t[\lambda_{it+1}Y_{it+1}]}{\lambda_{it}}. \tag{3}$$

[15] While I model inputs as being applied at a single stage prior to the realization of shocks for simplicity, in practice inputs are applied at different points in the growing season. I also discuss assumptions on the evolution of A_{it} across periods in Section 4.

[16] φ_{it+1} can be thought of as $e^{\phi_{it+1}}/\mathbb{E}_t[e^{\phi}_{it+1}]$ with ϕ_{it+1} a mean-zero shock from a known distribution. This makes φ_{it+1}, the unanticipated component of productivity, mean 1 by construction, with $\mathbb{E}_t[e^{\phi}_{it+1}]$ absorbed into A_{it} since it is known. I return to this in Section 4 when I discuss estimation.

[17] For simplicity, I assume that the law of one price holds in goods markets, which is consistent with the minimal cross-sectional variation in food prices within a township in the data.

where w_{ikt} is the (shadow) price of input k, which absent complete factor markets may vary across households i. Without loss of generality, I let $w_{ikt} = \bar{w}_{vkt}\tau_{ikt}$ in which $\bar{w}_{vkt}$ is the market price of input k in village v and τ_{ikt} is a wedge capturing deviations from the market price for household i. These include explicit taxes or subsidies, transaction costs, and binding quantity constraints. When $\tau_{ikt} > 1$ ($\tau_{ikt} < 1$) the household acts as if it faces a tax (subsidy) relative to the market price.

Since $Y_{it+1} = \mathbb{E}_t[Y_{it+1}]\varphi_{it+1}$, (3) can be rewritten to illustrate the distinct distortions more clearly as,

$$q_{ikt}\bar{w}_{vkt}\tau_{ikt} = \delta\alpha_k\mathbb{E}_t[Y_{it+1}]\Lambda_{it}, \quad \Lambda_{it} \equiv \frac{\mathbb{E}_t[\lambda_{it+1}\varphi_{it+1}]}{\lambda_{it}}, \quad \mathbb{E}_t[Y_{it+1}] = A_{it}\prod_k q_{ikt}^{\alpha_k}. \tag{4}$$

Λ_{it} captures how credit constraints and uninsured risk distort the intertemporal tradeoff between expenditure at planting and consumption at harvest. This can be decomposed into two terms. The first, $\mathbb{E}_t[\lambda_{it+1}]/\lambda_{it}$, reflects credit constraints: when households cannot borrow against future harvests, marginal utility at planting exceeds expected marginal utility at harvest. The second reflects uninsured risk, in which consumption at harvest depends on production shocks, making $\text{cov}_t(\lambda_{it+1}, \varphi_{it+1})$ potentially non-zero. Note that these two wedges, τ_{ikt} and Λ_{it}, are the only ways in which input demands differ from the perfect markets benchmark. Under a homothetic production function like Cobb-Douglas, Λ_{it} does not vary across inputs—financial frictions act as a *scale* distortion while input frictions act as a *composition* distortion.[18] However, the framework in Section 4 can accommodate more general production functions, as long as their input demands produce a well-identified Euler-like equation. As an example, I also extend this framework to a non-homothetic production function featuring differentially risky inputs in Appendix B, following Just and Pope (1978,

[18] In (4) the same Λ_{it} applies to all inputs chosen at period t, which simplifies the exposition and intuition. However, in the data I observe monthly input use and consumption. In the estimation procedure described in Section 4, I therefore divide t into monthly sub-periods and use the estimates of λ_{it} corresponding to the month that each input is applied. This allows a different stochastic discount factor to enter input demands at different points in a season.

1979), which allows financial frictions to also distort input composition.

3.2 Equilibrium

I now show how this model of farm-household distortions maps to aggregate misallocation.

Let $\eta \equiv \frac{1}{1-\gamma}$, which is a nonlinear transformation of returns to scale that approaches ∞ as production approaches CRS. In what follows, I drop time subscripts to ease notation. A decentralized allocation yields the following expression for the share of factor k in a given location allocated to household i. [19]

$$\omega_{ik} \equiv \frac{\frac{1}{\tau_{ik}} \left(A_i \Lambda_i \prod_l \tau_{il}^{-\alpha_l}\right)^{\eta}}{\sum_h \frac{1}{\tau_{hk}} \left(A_h \Lambda_h \prod_l \tau_{hl}^{-\alpha_l}\right)^{\eta}} \tag{5}$$

(5) is obtained by aggregating household first-order conditions (28) and implies that any allocation can be defined as a function of technology α, household TFP A, and distortions Λ and τ. It shows that a household's share of input k depends on three objects: its productivity A, its financial wedge Λ and its input wedges τ_k for each k, whose product is raised to the elasticity η. This also implies that when there are no distortions (i.e. $\Lambda_i = \tau_{ik} = 1$ for all inputs and households), the optimal allocation is

$$\omega_i^* \equiv \frac{A_i^{\eta}}{\sum_h A_h^{\eta}} \tag{6}$$

In this case, each input is allocated proportionally to household TFP, transformed by returns to scale.[20]

(5) shows how financial and input distortions jointly determine the allocation. A household with a low Λ_i (severe financial constraints) produces at an inefficiently small scale, but this can be either amplified or offset by input wedges If it faces $\tau_{ik} > 1$ for some input k, (effectively taxed in the market for that input), both distortions push in the same direction and

[19] Note that anything constant across households within the same location — such as the discount factor, market price of inputs, their aggregate supply and the market interest rate — cancel out of (5) .

[20] This is a standard result in the misallocation literature.

compound each other. But if $\tau_{ik} < 1$, then the effective subsidy partially offsets the financial friction. This is why the effect of reducing a single distortion in isolation is ambiguous — what matters for allocative efficiency is not whether it shrinks the distortions towards 1 but whether it pushes ω_{ik} closer to — or farther from — ω_i^*. One therefore must estimate A, τ, Λ and α to resolve this ambiguity.

4 Estimation Framework

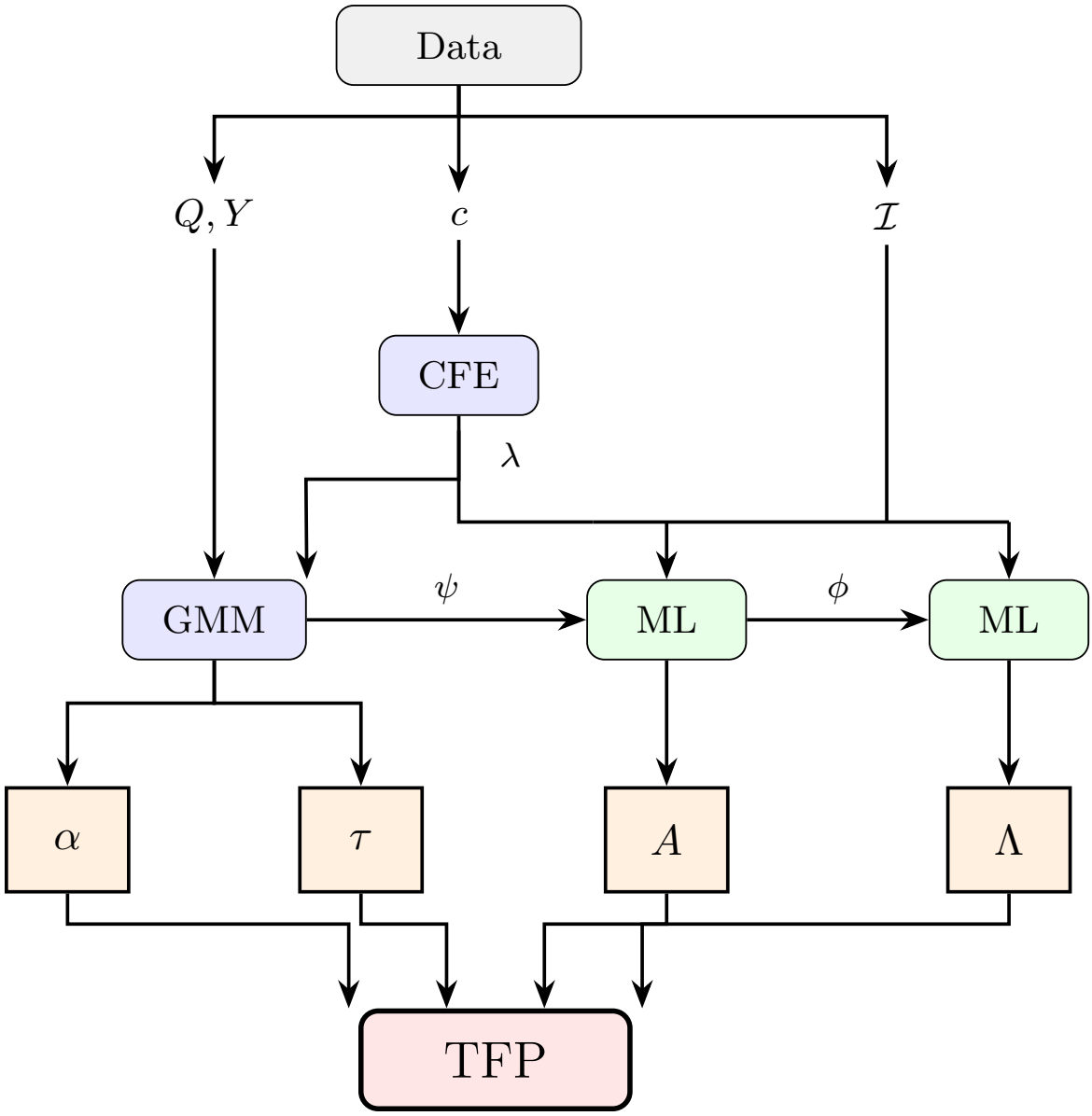
I now describe how to estimate the four main objects that determine misallocation: Technology parameters (α), households' anticipated TFP (A), financial wedges (Λ), and input wedges (τ). I proceed in two steps. First I use GMM to jointly estimate α and a parameterized version of τ from the production function and input demand moment conditions. The second step recovers A and Λ by isolating the ex-ante predictable components of realized TFP and consumption.

With estimates of α in hand, I can construct the ex post realization of TFP and its covariance with consumption. But recall from Section 3 that A and Λ are households' ex ante *expectations* of these objects. In a second step, I therefore project the realized values of TFP and $\frac{\lambda_{it+1}}{\lambda_{it}}\varphi_{it+1}$ onto the rich set of households' financial variables to isolate their anticipated components from realizations of ex post shocks, which cannot be considered misallocation (Gollin & Udry, 2021). The predicted values are my estimates of A and Λ, which I use in the decomposition exercise in Section 5.

4.1 Parameter Estimation (α and τ)

The model implies that both the physical production function and the first-order conditions for input demands must hold. The production function relates realized (ex post) output to inputs while first-order conditions relate input expenditure to (ex ante) expectations over future consumption and marginal products. I now estimate the technology parameters α

Figure 1: Estimation Flowchart



This flowchart illustrates the estimation procedure. In the first step data on inputs Q, output y and the marginal utility of expenditure λ are used to estimate the production coefficients α and the input frictions τ from the moment conditions described in Section 4.1. In the main analysis, λ is itself estimated from disaggregate consumption data using the CFE demand system in Ligon (2020, 2025), but this step is bypassed when assuming CRRA preferences in supplementary analyses. In a second step, I decompose the production function TFP residual ψ into an anticipated component A and an ex-post shock ϕ using machine learning with features from households' ex ante information sets, $\mathcal{I}$. I then further use machine learning to predict $\frac{\lambda_{t+1}}{\lambda_t}e^{\phi}$, which recovers financial wedges Λ. Together, this provides estimates of α, τ, A and Λ, needed to construct TFP following (5).

and a parameterized version of input wedges τ via GMM using these as moment conditions. While I estimate them jointly, I first describe each of the moment conditions in turn.

4.1.1 Production Function

I begin with the production function, which identifies the output elasticities α. Taking logs (denoted by lowercase letters) of the Cobb-Douglas specification (1) and first-differencing to eliminate household fixed-effects yields:

$$\Delta y_{it} = \sum_k \alpha_k \Delta q_{ikt} + \Delta \delta_{vt} + \Delta a_{it} + \Delta \phi_{it}, \tag{7}$$

where Δx_t denotes the difference between x_t and x_{t-1} and δ_{vt} is a location-year fixed effect. For the main analysis, I assume a_{it} follows a random walk following Anderson and Hsiao (1981, 1982) and similar applications in Shenoy (2017) and Manysheva (2021). This means that $\Delta a_{it} = \xi_{it}$, which is unpredictable when time inputs are chosen in $t-1$. Smallholder agriculture features few long-run investments in unobserved factors and future years' weather is hard to predict. In a robustness check, I relax this assumption and allow for productivity growth to follow a general Markov process.

Following Anderson and Hsiao (1981, 1982), lagged inputs q_{ikt-1} are valid instruments for Δq_{ikt} if they are orthogonal to ξ_{it} (holds by the random walk assumption) and $\Delta \phi_{it}$ (which holds by their definition as unanticipated shocks), as long as they are correlated (conditional on the location-year fixed effects). This relevance condition holds only if there are frictions preventing households from fully updating inputs in response to realizations of ξ_{it}.[21] Formally:

[21] For example, this would hold if households that use less land are less likely to rent additional land in response to a positive productivity shock. See Shenoy (2021) for a more detailed discussion.

$$
\begin{aligned}
&1.\mathbb{E}[q_{it-1}\Delta q_{it}] \neq 0 \\
&2.\mathbb{E}_{t-1}[q_{it-1}(\Delta y_{it} - \sum_k \alpha_k \Delta q_{ikt} - \Delta\delta_{vt})] = 0.
\end{aligned} \tag{8}
$$

Under these assumptions, one obtains the following moment condition:

$$
\mathbb{E}\left[q_{it-1}\left(\Delta y_{it} - \sum_k \alpha_k \Delta q_{ikt} - g(z_{it-1}) - \Delta\delta_{vt}\right)\right] = 0 \tag{9}
$$

The random walk assumption may be plausible in smallholder agriculture, which features few long-run investments in unobserved factors and unpredictable future weather. Subsequent results in subsubsection 4.2.1 showing that variables in households' pre-planting information sets predict minimal idiosyncratic variation in realized TFP lend support to this claim. I relax it in favor of a more general Markov assumption in Section E.2.

4.1.2 Euler Equation

The other set of moment conditions is the Euler-like equation for input investment, which identifies the τs and Λ under rational expectations. Under complete markets, households would simply set the (expected) expenditure shares of output for each input k to α_k, allowing us to simply calibrate α_k as observed expenditure shares. However, in general, output in each state of the world is weighted by a stochastic discount factor and the law of one price for inputs may not hold. These deviations from the perfect market benchmark are captured through the Λ and the τ, which we estimate to make the first order condition for input demands hold.

Estimating marginal utilities (λ) The financial wedge Λ depends on marginal utilities of consumption at planting (λ_t) and at harvest (λ_{t+1}). I estimate these λs from disaggregated expenditure data on 47 food and non-durable goods observed monthly over the 16-year panel, using the Constant Frisch Elasticity (Ligon, 2020, 2025) demand system. This is a flexible

demand system accommodates highly nonlinear Engel curves and nests many commonly used demand systems in the literature. Under CFE expenditures on good j take the following form:

$$\mathbb{E}[\log x_{ijt} - \alpha_j(p) + d(z) + \beta_j \log \lambda_{it}] = 0 \tag{10}$$

where p is the vector of prices, z is a vector of household characteristics, and β_j is the elasticity of expenditure on good j with respect to the marginal utility of expenditure λ_{it}. This structure permits the identification and estimation of the λ_{it} up to an exponent ζ that I estimate jointly with the input demand parameters,[22] I estimate λ using the full 196-month panel, allowing demands to vary with household composition as measured by the counts of members in different age-sex bins. Further details of the estimation procedure are in Appendix D.

Estimating τs With estimates of λ and α and observations of input quantities q and market prices $\bar{w}$, all that is left to estimate from (4) are τ, δ, and the nuisance parameter ζ. For δ, I convert Kaboski and Townsend (2011)'s estimate of an annual discount factor of 0.926 — also using the Townsend Thai Data — to reflect the number of months between the application of input k and harvest.[23]

Now I turn to estimating the τs (and ζ) via GMM. In practice τ_{ikt} needs to be parameterized if it is allowed to be time-varying.[24] This is because there are as many τ_{ikt}s as there are observations in the data and the residual from (11) would pick up any ex-post shocks to λ_{it+1} and Y_{it+1} as well as measurement error. In order to ensure that the estimates of τ_{ikt} pick up only variation that looks like an input distortion, I parameterize τ_{ikt} as a log-linear function of predetermined variables X_{ikt}, which are included in the instrument set Z_{ikt}. lagged input shares and household demographics (i.e. the counts of male and female adults and children).

[22] ζ is what Frisch (1959) calls "money flexibility" and will in practice also absorb sampling variation in the estimates of λ.

[23] I also show robustness to assuming an annual discount factor of 0.95.

[24] An alternative would be to rearrange (11) so that τ_{ikt} is in the residual and estimate a time-invariant $\hat{\tau}_{ik}$ as the average residual for each household-input. However, this would require both consumption shocks and input measurements to average out to zero in a $T = 16$ panel, potentially leading to drastic overestimates.

The former reflects that fact that input distortions are likely to be serially correlated while the latter reflects the well-established correlation between labor intensity and household size (LaFave & Thomas, 2016).

For instruments Z_{it}, I include lags of λ_{it} from the five months before input k is applied and the lagged value of the input from the prior season's production, along with a constant. The corresponding population moment condition becomes:

$$\mathbb{E}\left[Z_{ikt}(\bar{w}_{kvt}Q_{ikt}\tau_k(Z_{ikt}) - \alpha_k\delta\left(\frac{\lambda_{it+1}}{\lambda_{it}}\right)^{\zeta}Y_{it+1})\right] = 0 \tag{11}$$

where

$$\log\tau_k(Z_{ikt}) = \rho_{0k} + \rho_{1k}(\bar{w}_{kvt}Q_{ikt-1}/\sum_l \bar{w}_{lvt}Q_{ilt-1}) + \rho_{2k}\text{AdultMales}_{it} + \rho_{3k}\text{AdultFemales}_{it} + \rho_{4k}\text{ChildMales}_{it} + \rho_{5k}\text{ChildFemales} + \nu_{ikt}. \tag{12}$$

Why does this work? (11) is an *un*-conditional moment condition (note the absence of the t subscript on the expectation operator indicating conditioning on the household's information set). Meanwhile, the expectation operator in the first order condition (4) represents households' *subjective* expectations over future production shocks. While we can't observe these without full knowledge of households' information and beliefs, if expectations are rational, then we know that they are correct — and thus equal to what we'd expected to observe in the data — *on average*.[25]

[25] A standard concern with GMM estimators under rational expectations is that they require T to be large for aggregate shocks to average out — the sample mean of (11) is a random variable that is zero in expectation but is not necessarily equal to zero in a given year. With small T, realized aggregate shocks may have non-zero correlation with the instruments, biasing the estimates. While 16 years of data is modest by macroeconomic standards, I show via Monte Carlo simulations in that the resulting finite-sample bias is likely negligible in my setting.

4.1.3 GMM Implementation and Results

I now discuss how I implement the estimation of the αs and τs. The only calibrated parameter is the discount factor δ. I set the annual discount factor equal to Kaboski and Townsend's 2011 estimate of 0.926. However, unlike in the model, in which a period roughly corresponds to a season, different inputs are applied with varying time until the harvest. I therefore convert the annual discount factor to reflect the number of months remaining until the actual harvest.[26]

There are 50 remaining parameters to be estimated from 91 moment conditions. These include seven moment conditions to identify the seven αs, and 12 instruments (constant, lagged input expenditure and share, four household composition variables and five lags of λ) to identify six ρs for each of the 7 inputs plus a common ζ.

Table 1 reports the estimates and standard errors for the αs and their sum γ, in Column (1). The coefficients each take reasonable values for agricultural production functions, as does returns to scale $\hat{\gamma} = 0.83$, which implies an η of 5.95. The estimates of the ρ are not particularly notable, except for the negative coefficients on lagged input intensity for most inputs, which suggest persistence in the τs—households that used an input less intensively in previous seasons behave as if they currently face a higher price for it. I fail to reject the overidentifying restrictions of the model, with a J statistic of 39.40 ($p = 0.54$, 41 degrees of freedom).

The remaining columns of the table also report the estimates of ρ that are used to construct τ for each input, as well as the "money flexibility" parameter ζ. Note that $\hat{\rho}$ is only used to construct the τs used in the main analysis, while ζ is discarded. Histograms of each of the estimated $\hat{\tau}_k$ as well as the Cobb-Douglas price index $\prod_k \tau_{ikt}^{\alpha_k}$ are plotted in Figure C5. The most dispersion is observed for planting and harvest labor, while the least is observed for fertilizer. This aligns with the observed within township (tambon)-month price dispersion in each of these inputs (Table C4), in which the law of one price appears to hold for fertilizer

[26]Note that this assumes that households either know when they will harvest their crop, or that any uncertainty over how many months remain until harvest is uncorrelated with future consumption.

Table 1: GMM Parameter Estimates

	α	Const	Inp Intensity	M. Adults	F. Adults	M. Kids	F. Kids
Equip.	0.221***	0.637**	-0.195	-0.108**	0.118*	0.136**	0.168***
	(0.033)	(0.269)	(0.123)	(0.039)	(0.055)	(0.052)	(0.036)
Fert.	0.029*	-0.298	0.354***	-0.196***	0.199***	0.159***	0.270***
	(0.014)	(0.289)	(0.058)	(0.040)	(0.046)	(0.041)	(0.039)
Harv. Labor	0.128***	-1.290**	-0.144	0.341*	0.366	-0.025	-0.119
	(0.019)	(0.553)	(0.231)	(0.175)	(0.199)	(0.115)	(0.081)
Land	0.249***	0.349	-0.620***	-0.081*	0.120**	0.092*	0.132***
	(0.037)	(0.295)	(0.096)	(0.036)	(0.049)	(0.046)	(0.032)
Plant. Labor	0.072***	-0.053	-0.435***	-0.078	0.137**	0.109**	0.144**
	(0.015)	(0.371)	(0.062)	(0.044)	(0.050)	(0.043)	(0.046)
Seed	0.111***	0.364	-0.429***	-0.088**	0.121*	0.116**	0.132***
	(0.027)	(0.338)	(0.055)	(0.032)	(0.054)	(0.042)	(0.033)
Weed. Labor	0.022***	-1.369***	-0.337***	-0.184**	0.272***	-0.012	0.321***
	(0.006)	(0.390)	(0.068)	(0.063)	(0.079)	(0.075)	(0.055)
γ	0.832	ζ	2.682				
γ s.e.	(0.038)	ζ s.e.	(0.126)				
J	39.399						
p-val	0.542						

This table shows the results of the GMM estimation of α and the parameters ρ that make up τ. The estimation is carried out using the continuously updated GMM estimator of Hansen et al. (1996). Column (1) shows the estimates of the production function coefficients α while the remaining columns show the estimates of the coefficients ρ parameterizing τ for each input. The bottom of the table also reports the returns to scale parameter γ and money flexibility parameter ζ and their standard errors. The J statistic and p value from the test of the 42 overidentifying restrictions (92 moment conditions, 50 parameters) are also reported below. I fail to reject the null that all the moment restrictions are valid.

but not for other inputs.

4.2 Productivity and Financial Wedges (A and Λ)

The previous section exploited the structure of the model to identify α and τ from moment conditions derived from the farm-household model. However, identifying anticipated TFP A and the financial wedge Λ is more complicated, as it requires distinguishing between households ex-ante predictions over productivity and consumption from the realization of ex-post shocks—which the moment conditions in the previous step averaged out. I take a data-driven approach to this, in effect approximating the prediction problem the household

itself faces at planting time using flexible machine learning methods with a rich set of features, rather than placing additional parametric structure on households' expectations.

4.2.1 TFP

I recover the TFP residual from the production function estimates:

$$\psi_{it+1} \equiv y_{it+1} - \sum_k \hat{\alpha}_k q_{ikt} \tag{13}$$

and decompose it into a household fixed effect $\bar{A}_i$ plus deviations. Many papers in the misallocation literature assume that productivity is time-invariant within households—that is, all of ψ_{it+1} beyond the fixed effect is unanticipated—and stop here. This is a strong assumption, especially in a 16-year panel where households may experience persistent changes in soil quality, management ability, or local conditions.

I relax this assumption by projecting ψ_{it+1} onto variables in households' time-t information sets in order to distinguish the anticipated component a_{it} and the unanticipated component ϕ_{it+1}. If a component of TFP were truly unanticipated, it should be uncorrelated with variables known at planting. Conversely, any predictable component should show up in consumption and investment behavior—under incomplete markets, households respond to anticipated shocks by adjusting savings, borrowing, and input purchases.[27] I attribute the predicted component to anticipated TFP ($\hat{a}_{it} = \exp(\bar{a}_i + \hat{\xi}_{it})$) and the residual to the unanticipated shock $\hat{\phi}_{it+1}$. This ensures that $\hat{\phi}_{it+1}$ is mean-zero from the household's ex-ante perspective. This is similar in spirit to the use of $g(z_{it-1})$ to proxy for anticipated *changes to* productivity in (9), except I am now trying to predict the *level* of productivity at time t—in

[27] For example, if a household anticipated a positive TFP shock, it should adjust consumption (in a direction determined by the intertemporal elasticity of substitution) and, if possible, borrow to finance additional inputs. Under binding borrowing constraints, we might also expect substitution away from other investments. Note that this does not conflict with the identifying assumption for the production function moments because different periods' information sets are being considered. The production function assumes that time t shocks are unpredictable using the prior *year*'s information (implicit in the AR(1) assumption, which I fail to reject), while this procedure allows for some of *this year's* TFP shock to be known prior to planting (for example if the farmer receives a rainfall forecast or falls ill prior to planting.)

other words, I am trying to predict $h(a_{it-1})+\xi_{it}$ rather than $h(a_{it-1})-a_{it-1})$. Moreover, I am interested in the predictions themselves, rather than trying to control for them to estimate coefficients.

In the main results, I use gradient boosted trees (XGBoost) with five-fold cross-fitting to predict both the anticipated productivity innovation ξ_{it}, leaving $\hat{\phi}$ as the residual. This produces a higher R^2 than random forest or linear methods,. In the Appendix I show results using these methods and additional steps to prevent overfitting, which would lead to inflated estimates of financial misallocation. An important caveat is that I can only project onto *observed* variables in the information set. If households have private information that predicts productivity or marginal utility but is not captured in the survey, the "unanticipated" component $\hat{\varphi}$ will contain some forecastable variation, and $\hat{\Lambda}$ will be measured with error. This would attenuate the estimated contribution of financial frictions to misallocation, making my estimates conservative.[28]

After purging $\hat{\psi}_{it+1}$ of its time-average $\bar{a}_i$, the XGBoost model with 5-fold cross-fitting predicts $\hat{\xi}$ with an R^2 of 20.7%. Again, this suggests that households anticipate about one-fifth of the deviation in ex-post realized TFP from their average. This prediction is heavily spatially-and temporally correlated and increasing over time, consistent with it capturing secular trends in productivity.[29] Overall, this suggests that most of the variation in TFP is unanticipated by households at the time of planting.

4.2.2 Λ

The previous step allows me to construct an estimate of *realized* $\frac{\lambda_{it+1}}{\lambda_{it}}\varphi_{it+1}$, the household's *subjective time t expectation* of which is Λ_{it}. Rational expectations allowed us to identify the τs from (11), by guaranteeing that *on average* the realized value of $\frac{\lambda_{it+1}}{\lambda_{it}}\varphi_{it+1}$ equals Λ_{it}.

[28]Note that even if I perfectly observed households' information sets and knew their beliefs, I would not obtain an R^2 of 1 for any of these predictions, as some of the variation in the target variable is fundamentally unpredictable from the *household's* perspective, not just the econometrician's.

[29]When demeaning $\hat{\psi}_{it+1}$ by location-year, the gradient-boosted prediction has an R^2 of only 0.015. This suggests that there is very little *idiosyncratic* growth in anticipated TFP for households to predict, justifying the random walk assumption in (which is conditional on village-year fixed effects).

However, any given observation contains both the ex-ante Λ and the realization of ex-post shocks. Attributing these ex-post shocks would severely inflate estimates of misallocation (Gollin & Udry, 2021). Therefore, I need to isolate the predictable component of the realized value of $\frac{\lambda_{it+1}}{\lambda_{it}}\varphi_{it+1}$ in order to obtain a valid measure of financial distortions.

This is a formidable challenge since I don't observe households' subjective expectations of λ_{it+1} and how they anticipate it will respond to realizations of ϕ_{it+1}. This depends on households' wealth, investment portfolios, positions in insurance networks, and beliefs, which all almost certainly fluctuate over time. [30] In contrast, taking the time-series covariance of a household's realized production shocks with realized marginal utilities at harvests (as Samphantharak & Townsend, 2018, do with stochastic discount factors) would miss important differences in state variables. Furthermore, estimation error from the time-series covariance is not something that simply averages out when computing estimates of misallocation, since misallocation is a function of the (cross-sectional) variance of Λ_{it}. Therefore, I use cross-validated machine learning to capture the dependence of Λ_{it} on the underlying state variables as flexibly as possible, rather than making additional assumptions and modeling it structurally.

This is not without caveats. In particular, if the machine learning model cannot fully capture households information sets, most predictors will tend to understate financial misallocation, since the variance of the estimated $\hat{\Lambda}_{it}$ would be less than that of the true Λ_{it}. Moreover, a learner that minimizes mean-square error will also further attenuate the variance of $\hat{\Lambda}$ at the cost of introducing a tolerable amount of bias in the prediction.[31] Together, these make estimates of financial misallocation somewhat conservative. In contrast, the naive time-series approach, which uses household-level covariances without passing through a prediction

[30] For example, under CRRA utility with this would be expressed as

$$\mathbb{E}_t[\lambda_{it+1}\varphi_{it+1}] = \int_{\varphi} \frac{\varphi}{(R_{it+1}(\varphi)B_{it} - B_{it+1}(\varphi) + A_{it}\varphi\prod_k q_{ikt}^{\alpha_k} - \sum_k w_{ikt+1}(\varphi)q_{ikt+1}(\varphi))^{\theta}} df(\varphi)$$

, where returns to assets B and choice variables at $t+1$ may depend on the realization of φ.

[31] Moreover, the bias-variance tradeoff that machine learning entails conflicts with the rational expectations assumption in subsection 4.1, which assumes that households make *unbiased* predictions about future shocks.\

step, provides a benchmark without this built-in conservativeness despite not capturing the model-consistent expectation.

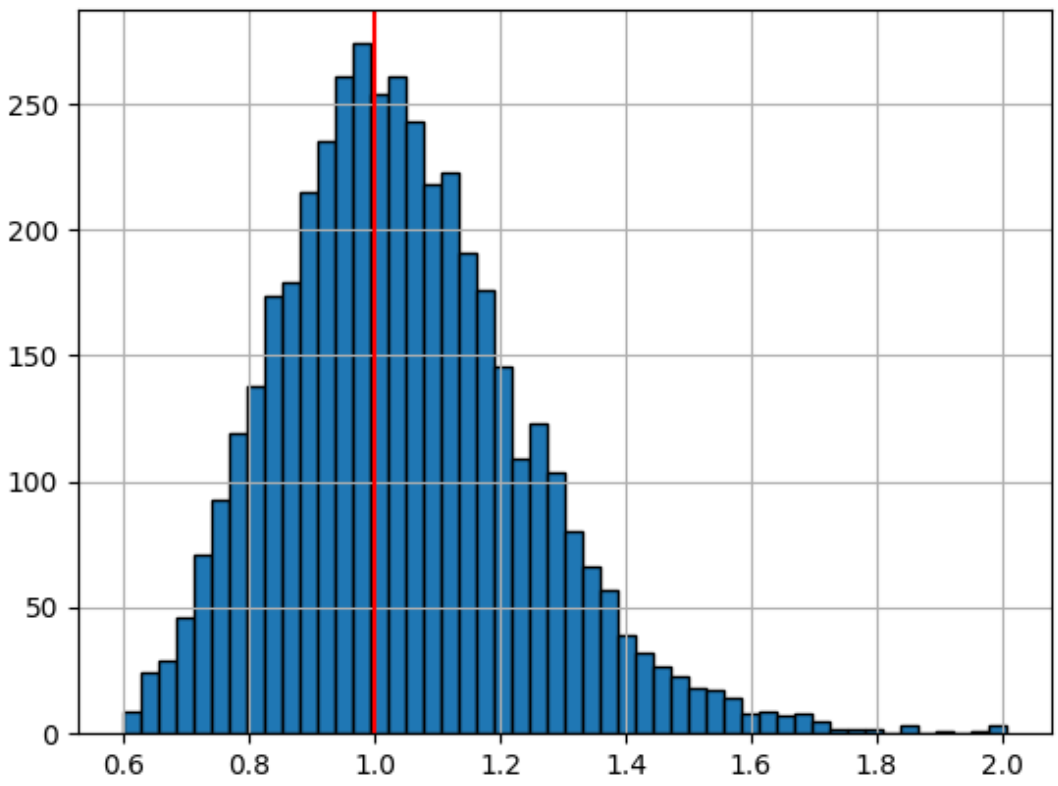
To implement this, I predict the realized value of $\lambda_{it+1}\varphi_{it+1}$ using a rich set of features from households' time t information sets. Given that $\mathbb{E}_t[\varphi_{it+1}] = 1$ by construction, any variation in $\hat{\Lambda}_{it}$ reflects either a predictable wedge between marginal utilities at planting and harvest (credit market imperfections) or the dependence of λ_{it+1} on φ_{it+1} (uninsured risk).

The gradient booster predicts the realized $\frac{\lambda_{it+1}}{\lambda_{it}}\varphi_{it+1}$ with an R^2 of 0.28, suggesting that households anticipate that future production shocks will cause a great deal of variance in consumption growth. Naturally, this distribution of the *predicted* $\frac{\lambda_{it+1}}{\lambda_{it}}\varphi_{it+1}$ is a shrunken version of the distribution of the *realized* $\frac{\lambda_{it+1}}{\lambda_{it}}\varphi_{it+1}$, as seen in Figure C4. Figure 2 shows the distribution of estimated financial wedges $\hat{\Lambda}$. The mean (median) of $\hat{\Lambda}$ is 0.945 (0.928) and the distribution is fairly tight, with a standard deviation of 0.18 and an interquartile range of 0.82–1.05. Given that Λ enters input demands raised to the elasticity $\eta \approx 5.95$ (5), this suggests the median household is only producing at about 64% of its desired scale due to credit constraints and risk, given prices and input wedges.

Note that about one-third of households have a $\hat{\Lambda} > 1$, which suggests that agriculture is a hedge against the rest of the household's portfolio. This is indeed most prevalent in the wealthier and more urbanized province Chachoengsao, where agriculture makes up a smaller share of household's income. Together, this is consistent with other evidence of non-missing but incomplete financial markets from the Townsend Thai Data (e.g. Kaboski & Townsend, 2011; Karaivanov & Townsend, 2014; Samphantharak & Townsend, 2018; Kinnan et al., 2024).

Extreme gradient boosting (XGBoost) is my preferred specification, as it achieves a much higher cross-fitted R^2 than random forest. Theoretically, it is also better suited for this problem than random forest, which has a tendency to compress predictions and thus overly attenuate the variance of $\hat{\Lambda}$. In (Figure E1), I show the predictions of $\hat{\Lambda}_{it}$ from XGBoost, Random Forest, and the time-series covariance, alongside the empirical distribution of $\frac{\lambda_{it+1}}{\lambda_{it}}\varphi_{it+1}$. The

Figure 2: Histogram of Estimated Financial Distortions $\hat{\Lambda}$



The figure shows the distribution of the estimated financial distortions $\hat{\Lambda}_{it}$ from the main gradient-boosting specification. The red line at 1 corresponds to the benchmark of no financial frictions. The mean (median) of Λ_{it} is 0.945 (0.928).

distribution of all three estimates of $\hat{\Lambda}_{it}$ is much tighter than the raw values of $\frac{\lambda_{it+1}}{\lambda_{it}}\varphi_{it+1}$, reflecting the considerable degree of variation due to the realization of ex-post shocks. Among the three estimates, the random forest is the most compressed, while the XGBoost and time-series predictions have similar marginal distributions. This is perhaps reassuring that shrinkage inherent in XGBoost does not substantially attenuate the estimated variance of financial frictions. Nevertheless, one can think of the results produced by the XGBoost and the naive specification as plausible bounds on financial misallocation.

5 Decomposition of Misallocation

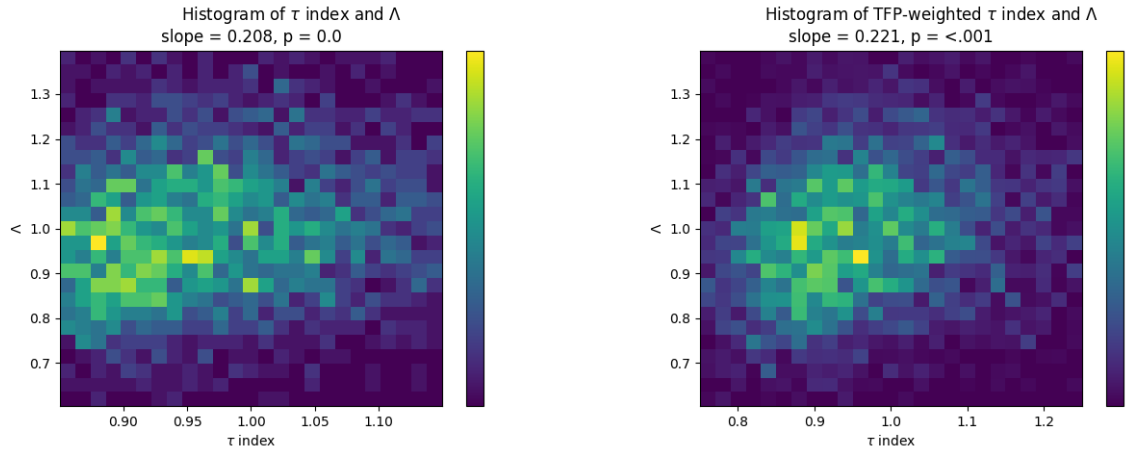
While the results of the previous section showed the *marginal* distributions of $\hat{\Lambda}$, $\hat{\tau}$ and $\hat{A}$, what matters for aggregate productivity and misallocation, according to Section 3 is their *joint* distribution.[32] Before delving into the decomposition of misallocation, I provide some descriptive graphical evidence to characterize this distribution.

5.1 Descriptive Results

Figure 3 plots 2D histograms of input and financial distortions and reports their correlation coefficients. The left panel plots the unweighted Cobb-Douglas price index of τs, $\prod_l \tau_{ilt}^{\alpha_l}$ against the estimates of financial distortions Λ, while the right panel plots each weighted by TFP A. The positive correlation between τ and Λ suggests that, on average, more financially constrained households (low Λ) are relatively *subsidized* on inputs (low τ). This corresponds to the conventional wisdom that poorer households oversupply labor to their own farms under imperfect labor markets (Breza et al., 2021; Jones et al., 2022; LaFave & Thomas, 2016). This correlation is even stronger when weighting by TFP, which reflects how distortions for more productive households exert greater influence in equilibrium, since these households command larger shares of total inputs .

[32]This is an extension of results in Hsieh and Klenow (2009) and Adamopoulos et al. (2022) regarding the covariance between wedges as a sufficient statistic for misallocation.

Figure 3: Joint distribution of τ and Λ



This figure plots histograms of Λ and τ in 25×25 bins and reports the slope coefficients. The left panel plots the unweighted joint distribution of the Cobb-Douglas price index of τs, $\prod_l \tau_{ilt}^{\alpha_l}$, against the estimates of financial distortions Λ. The right panel plots the TFP-weighted joint distribution, which corresponds to the model-consistent moment (see 5).

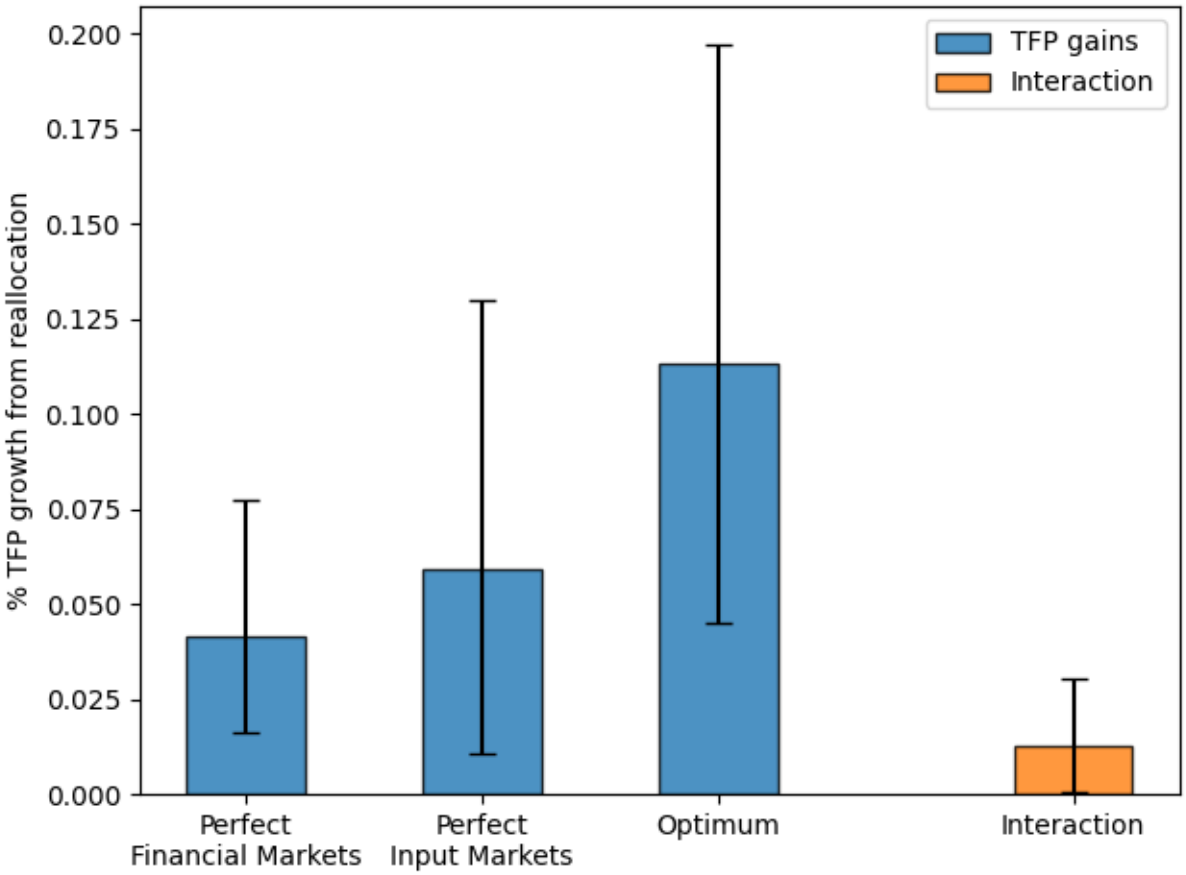
This implies that the observed distortions partially offset each other — relaxing credit constraints would disproportionately direct capital toward farms that are effectively subsidized on inputs. The direct gains from relaxing credit constraints are large enough to swamp this effect but are smaller than they would be if credit constraints were uncorrelated with input distortions.[33] The results also show that distortions for land and labor are positively correlated. Most of the misallocation literature rules this out by assumption, modeling τ as a distortion in the *relative* price of land and labor. However, I am able to separately identify *absolute* price wedges by including consumption moments in the estimation.

5.2 Main Decomposition

I now decompose the productivity dispersion in the Thai data into four parts: misallocation due to imperfect financial markets, misallocation due to imperfect factor markets, their interaction effects, and a residual that cannot be interpreted as misallocation according to the model. I do this by computing *expected* aggregate productivity under the following four

[33] TFP governs the incidence of these distortions; since it is the sole determinant of scale under the efficient allocation, multiplicative wedges such as Λ or τ exert a large influence on the aggregate economy when it affects firms that command more inputs.

Figure 4: Decomposition of TFP gains from reallocation



The figure decomposes TFP gains from reallocation, summed across years as a percentage of status quo aggregate TFP. The first three bars show gains under perfect financial markets, perfect input markets, and full efficiency, respectively. The fourth (orange) bar shows the interaction effect: the difference between the full-efficiency gain and the sum of the two partial gains. Aggregate supply of all inputs is held fixed at the township level. Results are computed using CFE demands, restricting the sample to rice plots, and aggregating to the farm level. 90% confidence intervals from 200 bootstrap replications are shown.

scenarios: (1) the first-best allocation; (2) the baseline allocation, with all of the distortions I measure; (3) an allocation with perfect financial markets and the observed input wedges; (4) an allocation with perfect input markets and the observed financial wedge. I compute aggregate TFP under these allocations within township-years and then sum up these gains across townships in each of the 16 years of the sample.

5.2.1 Baseline Misallocation

Figure 4 plots the TFP gains from reallocation under three scenarios as a percentage of (expected) aggregate TFP in the observed allocation: (1) eliminating financial distortions (i.e. setting $\Lambda = 1$) holding input frictions τ fixed; (2) eliminating input distortions (setting $\tau = 1$) while holding Λ fixed; and (3) eliminating all distortions. The difference between (1)+(2) and (3) corresponds to the interaction between input and financial misallocation, which can be positive or negative. I hold aggregate supply of all inputs fixed at the township level, so that TFP is directly proportional to aggregate output.[34] Confidence intervals from 200 bootstrap replications are shown.

The TFP gains from full reallocation are 11.3%. At face value, these baseline estimates are similar to Shenoy (2017)'s estimates from Thailand, which I discuss below. On the other hand, my results are an order of magnitude lower than some estimates from Africa of up to 286% gains from reallocation Aragon et al. (2022) and Chen et al. (2023).

It is clear from the first two bars in Figure 4 that both sets of markets contribute significantly to misallocation in isolation. Perfecting financial markets while holding observed input distortions intact achieves about 36.5% of the possible efficiency gains, or 4.1% of observed TFP. Similarly, removing input distortions holding observed financial frictions intact achieves about 52.4% of these gains (5.9% of TFP).

Notably, these two partial gains sum to less than 100% of the total gains, meaning the gains from full reallocation are more than the sum of its parts. The orange bar in Figure 4 shows this interaction effect directly: the additional gain from jointly eliminating both distortions beyond what is achieved by removing each in isolation. This positive interaction arises because Λ and τ are positively correlated (when weighted by TFP). In other words, the most financially constrained households are relatively subsidized in input markets, especially labor, as shown in Figure 3.[35] The effect of relaxing financial constraints is thus attenuated — but

[34] Allowing tradable inputs (fertilizer, seed, and equipment) to be imported at constant prices yields larger output gains (18.8% under the efficient allocation) but does not affect TFP gains, since these are determined solely by the reallocation of inputs across producers.

[35] This reflects the common finding that poorer households tend to oversupply labor to their own plots.

Table 2: Decomposition of Gains by Input Market

	With Financial Constraints	Perfect Financial Markets
Equip.	0.014	0.056
Fert.	0.004	0.046
Harv. Labor	0.021	0.063
Land	0.014	0.059
Plant. Labor	0.005	0.049
Seed	0.008	0.051
Weed. Labor	0.003	0.045

This table shows the gains from removing distortions τ_{ikt} in individual input markets, both with the observed financial constraints and under perfect financial markets. This is shown for the closed economy case, CFE demands, and restricting the sample to rice crops at the farm level.

not offset — by reallocating resources to farms made inefficiently large by other distortions. Overall, these patterns suggest that the effects of policies targeting a single market failure would be attenuated, rather than amplified, by failures in other markets. In contrast, policies addressing multiple market failures would have positive complementarities.

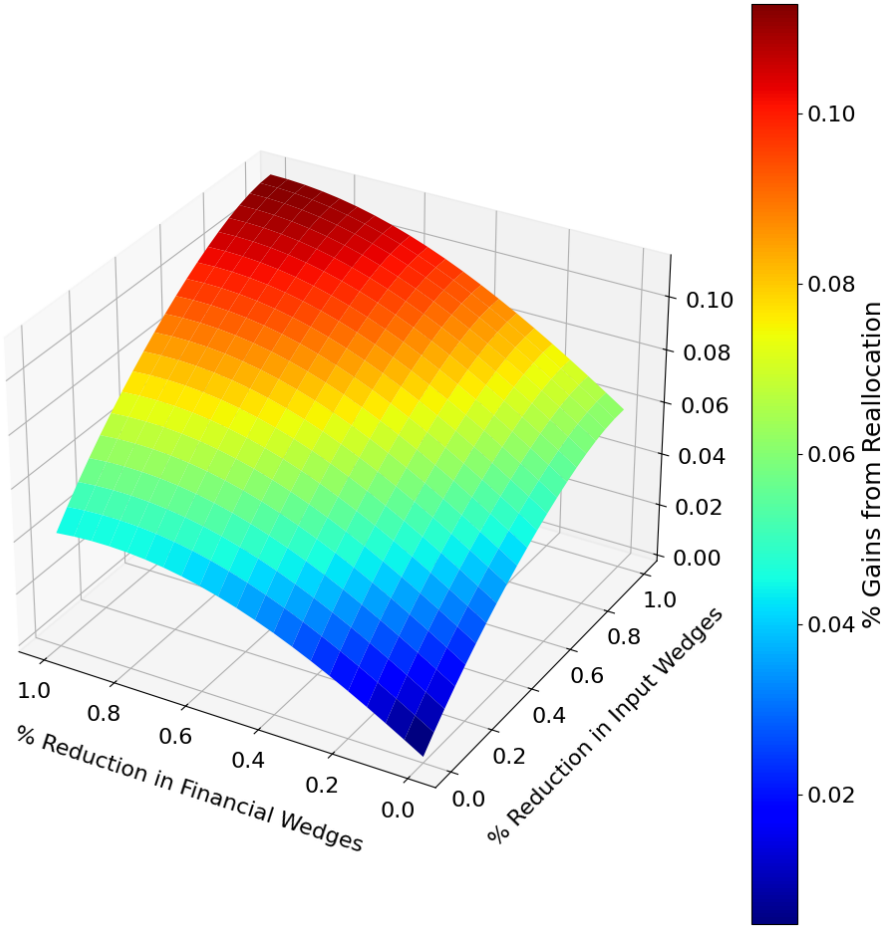
I further decompose factor misallocation into that resulting from frictions land, equipment, and total labor as well as labor at different points in the season. Table 2 shows the results of removing wedges from each of these markets, with and without relaxing financial constraints. Reducing frictions in labor markets is slightly more effective than for land markets, despite them accounting for roughly equal expenditure shares. The sum of gains from reducing individual frictions is also more than the gains from reducing all of them simultaneously. While input frictions are negatively correlated with financial distortions, they are positively correlated with each other. In other words, reducing frictions in land markets also indirectly addresses labor market distortions by reallocating resources toward households that are relatively taxed.

5.2.2 Intermediate Policies

The results above consider the gains from completely eliminating one set of distortions while holding others fixed at observed values. However, policymakers likely have a menu of policy

instruments to choose from, but may not be able to fully eliminate distortions. The model allows me to estimate aggregate TFP under any values of Λ and τ. I therefore conduct a simple illustrative exercise in Figure 5, in which I plot the TFP gains from uniform partial reductions in τs and Λs. This approximates the marginal returns to reductions in distortions. However, modeling the effects of a specific policy would require assumptions on the specific institutions underlying the distortions I measure, which also govern the second-order effects of how a change in τ affects Λ (and vice versa).

Figure 5: Gains from partial reductions of τ and Λ



The figure shows the decomposition of misallocation using the TFP-based measure under different reductions of input and financial wedges. I compute aggregate TFP under each scenario shrinking Λ and τ towards unity by increments of .05. The origin corresponds to the status quo allocation and (1,1) corresponds to the efficient allocation. The vertical axis shows the percent increase in aggregate TFP relative to the status quo allocation. The figure uses CFE demands and restricts the sample to rice plots, aggregating to the farm level.

Figure 5 illustrates the complementarities between policies that reduce both sets of dis-

tortions. In particular, it shows that the marginal returns to reducing either set of distortions are initially high, as evidenced by the steep slope near the origin, but sharply reducing. This implies that small improvements to input and financial markets together could achieve much of the attainable productivity gains, more so than focus on a single set of distortions in isolation. Moreover, if one knew the relative costs of reducing each set of distortions, the gradient of Figure 5 would define an expansion path for the social planner in terms of which distortions to target as its budget shifts out. Additionally, Figure 5 shows that these marginal returns are not monotonic: at baseline levels of input (financial) distortions, going from 10% of observed financial (input) distortions to perfect financial (input) markets actually worsens efficiency.

5.3 Methodological Differences and Measurement Error

I now describe how estimating both Λ and τ helps alleviate concerns about measurement error. With both Λ and τ, aggregate productivity under a given allocation can be computed in two ways: taking the observed allocation and then "removing" a distortion or taking the first-best allocation and "adding a distortion". To see this, note that the efficient allocation (6), which is just A_{it}^{η} over its sum, can also be written as a function of observed input demands and wedges by inverting (4) and dividing out constants:

$$\omega_{it}^{*} = \frac{Q_{ikt}\tau_{ikt}\left(\prod_{l}\tau_{ilt}^{\alpha_l}\Lambda_{it}\right)^{-\eta}}{\sum_{h=1}^{N} Q_{hkt}\tau_{hkt}\left(\prod_{l}\tau_{hlt}^{\alpha_l}\Lambda_{ht}\right)^{-\eta}}. \tag{14}$$

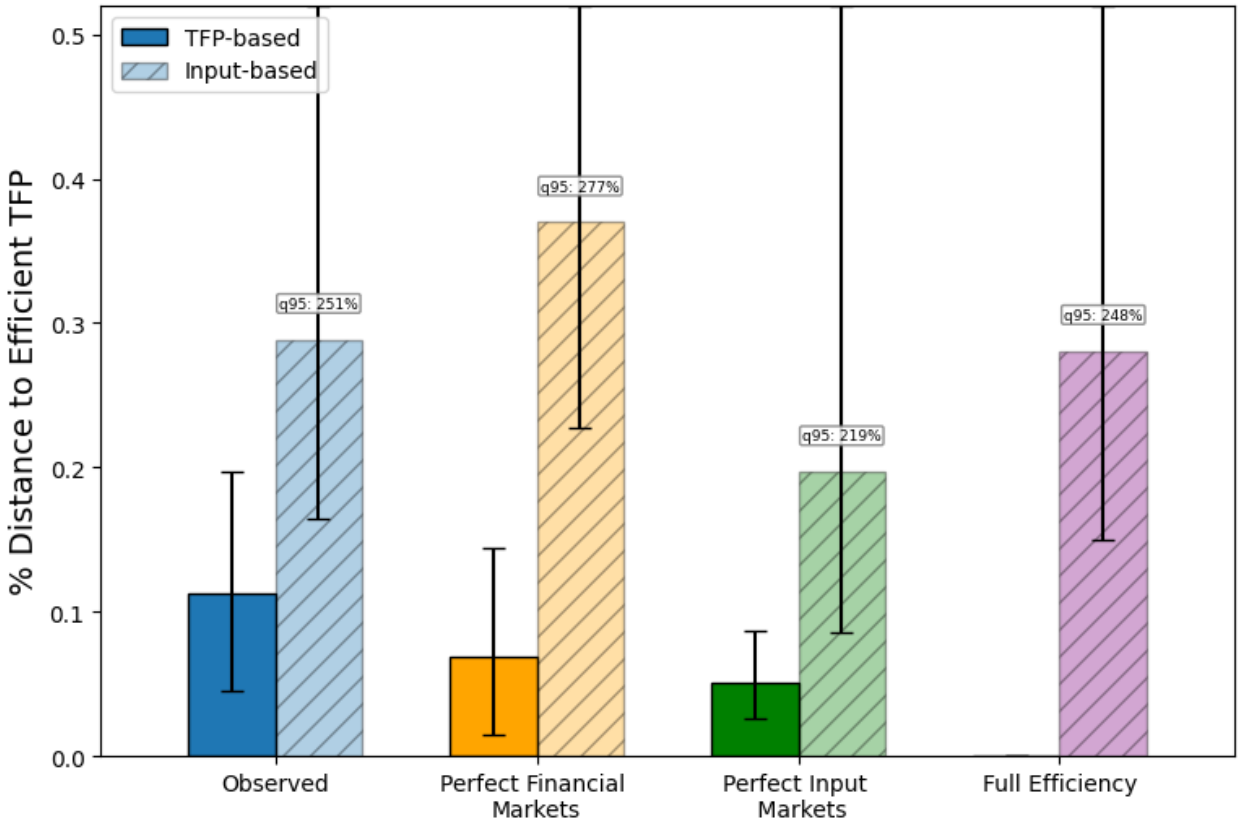
Likewise, under the status quo, rewriting (5) should simply yield

$$\omega_{ikt} = \frac{Q_{ikt}}{\sum_{h=1}^{N} Q_{hkt}} \tag{15}$$

This allows me to compute TFP using either (5) or (14) and then aggregate for any values of Λ and τ. However, this requires estimates of both Λ and τ.

If inputs were measured perfectly and τ and Λ were estimated without error, then these

Figure 6: Aggregate TFP relative to optimum, model- vs. input-based measures



The figure shows the percentage of foregone attainable output from the four benchmark allocations considered (the observed allocation, efficient allocation, perfect financial markets with input wedges intact, and perfect input markets with financial wedges intact). The solid bars compute these using the TFP-based measure of misallocation, using (5). The shaded bars are calculated by taking raw input observed in the data and augmenting them by the estimated τ and Λ, where relevant. 90% confidence intervals from 200 bootstrap replications are plotted; where the upper bound (95th percentile) exceeds the limit of the y axis, the bound is printed instead. Results are computed using CFE demands, only rice plots, and aggregating to the farm level.

two approaches should produce identical estimates. The difference is that the former approach (5 and 6) uses estimated TFP while the latter (14 and 15) uses raw input measurements. Which estimate is preferable depends on how severe measurement error in inputs is relative to the errors in estimated objects. Given that estimates of TFP are less noisy than the raw inputs used to estimate them, one would therefore expect estimates using the TFP-based measures in (5) and (6) to be more reliable than the input-based measures in (14) and (15).

How different are the conclusions these measures produce in the data? To make this com-

parison, it will be useful to denominate misallocation by the *attainable* output (equivalent to TFP when aggregate input supply is fixed) forgone due to distortions in each scenario. Figure 6 compares the TFP-based results (solid bars) with the input-based results (shaded bars). The solid bars recast the estimates from Figure 4, showing the percent of attainable TFP foregone in the observed allocation (blue), with only input frictions (orange), and with only financial frictions (green). By definition, the optimum allocation achieves all the attainable TFP so there is no solid purple bar.

Now contrast these TFP-based results with the shaded bars, which are computed using the input-based measure. As discussed in Section 3, these two panels would yield identical results if there were no measurement error and the model was perfectly specified. However, the differences between the two panels are quite striking when comparing bars of the same color in Figure 6. First, measured misallocation in the status quo is similar when using the input-based rather than the TFP-based measure, albeit much noisier. Second, it appears that perfecting financial markets would nearly *double* misallocation, rather than decreasing it.

Most strikingly though, the implied "optimum" allocation is not only suboptimal but actually performs worse than the observed allocation.

How is this possible? Recall that decompositions using the input-based measure are computed by adding distortions to the observed allocation, which includes mismeasured inputs. The shaded green bar is calculated by equalizing factor ratios in a way that preserves scale across farmers: this is the model of an exchange economy that serves as a lower bound on factor misallocation in Shenoy (2017). The purple bar is then calculated by reweighting those demands by $1/\Lambda$, removing estimated financial frictions.[36] The extremely wide bootstrap confidence intervals on the input-based measures further illustrate this point: Unlike the TFP-based measures, these are highly sensitive to different samples , further underscoring

[36]Note that the same wedges are used in each set of results but for different specifications. Input wedges are used to compute the orange and blue solid bars and the green and purple bars in the right panel. Meanwhile, financial wedges are used to compute the blue and green solid bars and the purple and orange shaded bars.

their susceptibility to measurement error.[37]

The raw input-based estimates are higher across the board than those using only estimated components. The conflicting result that removing financial frictions would worsen misallocation can be explained by their negative correlation with input measurement error. In other words, measurement error looks like a distortion that is partially offset by financial frictions — removing financial wedges thus makes this spurious distortion appear worse.[38]

These differences between the TFP and input-based measures are quite robust across specifications. Together, these results underscore the importance of separately identifying both input and financial distortions. Without a credible estimate of financial distortions, one would need to rely on noisily measured inputs and arrive at qualitatively different conclusions about the effects of market interventions.

5.4 Alternative specifications and robustness checks

In this subsection, I explore the sensitivity of the results to different specifications and underlying assumptions. The results of these robustness checks I describe below are also are summarized in Table 3. The overall patterns of modest levels of misallocation, positive interactions from addressing multiple distortions, and much larger and less stable results from using the input-based rather than TFP-based measure hold across all specifications.

5.4.1 Financial Misallocation using Time-Series Covariance

As discussed in Section 4, the financial wedge estimates from the ML prediction step may be somewhat conservative. In Figure E2, I reproduce Figure 5 using the time-series median of the realized $\frac{\lambda_{it+1}}{\lambda_{it}}\varphi_{it+1}$ to construct a time-invariant Λ_i for each household instead, bypassing the attempt to predict them ex-ante using machine learning. While not a theory-consistent al-

[37]Note that the asymmetric bootstrap confidence intervals reflect that the estimand is strictly positive by construction and exponentiated.

[38]Arthi et al. (2018) find that labor inputs are more upwardly biased for smaller farms. Since Figure 3 shows that these households are more financially constrained, financial constraints would then be negatively correlated with the measurement bias. Relaxing these constraints would therefore allocate more resources to farms that appear artificially large in the raw data.

ternative, it may provide an informative, less-conservative benchmark. Estimates of financial misallocation are substantially larger (7.3% vs. 4.1%), as expected given the less conservative approach, and total misallocation rises to 14.7%. Input misallocation is slightly lower (4.5% vs. 5.9%), reflecting the larger role attributed to financial frictions. More importantly, the positive interaction effects from addressing both sets of distortions and diminishing marginal returns to addressing a single distortion are preserved. Thus, the qualitative conclusions don't appear to hinge on the method used to predict Λ_{it}.

5.4.2 Alternative Specifications

Table 3 summarizes the main decomposition results from a battery of alternative specifications including CRRA preferences instead of the more general CFE specification, classifying labor as a single input rather than disaggregating it by production stage, including non-rice crops in the sample, using observed market rents to price non-transacted plots rather than hedonically adjusting for land quality, double post-Lasso production function controls, and a naive (non-ML) moment estimator.

The main results are qualitatively robust across specifications, with estimates of total misallocation ranging from 7.8% to 17.1%. All specifications show small positive interaction effects from relaxing both sets of distortions, ranging from 0.3–2.9% and produce smaller estimates of total misallocation than the input-based specification. Notably, the input-based measure produces estimates of 569% total misallocation when including all crops in the estimation sample, although the TFP-based estimate remains stable at 7.8%. I interpret this counterfactual as a test of sensitivity to production function mis-specification: the optimal factor shares for these crops are almost certainly different from those of rice. The differences between the two estimates here can be attributed to a few households cultivating non-rice crops with extremely high $\hat{A}$. In the first-best benchmark , these households are allocated a massive share of factors and makes it appear that attainable productivity is unrealistically high. The input-based measure attributes any difference between this and TFP under the

baseline allocation as misallocation. In contrast, the TFP-measure backs out the status quo TFP by distorting the first-best allocation with the estimated wedges. Since the wedges are not correlated with the extreme $\hat{A}$ values from non-rice crops, the inconsistency of these A's with the observed factor allocation doesn't factor into the misallocation estimates. I therefore interpret this as the TFP-based estimate being more robust to production-function misspecification.

Table 3: Robustness of Main Results Across Specifications

	Baseline	CRRA	No Hedonic	All Crops	Labor Agg	Lasso	Naive
γ	0.832	0.839	0.842	0.853	0.831	0.850	—
	[0.759, 0.894]	[0.728, 0.898]	[0.804, 0.950]	[0.808, 0.931]	[0.768, 0.899]	[0.768, 0.917]	
J-stat (p)	39.4 (0.542)	96.7 (0.000)	48.4 (0.200)	35.3 (0.722)	16.9 (0.963)	54.9 (0.072)	—
Total misallocation (TFP)	11.3	17.1	12.8	7.8	9.5	12.9	14.7
	[4.5, 19.7]	[12.1, 89.1]	[3.3, 34.0]	[3.3, 12.7]	[5.1, 35.4]	[5.0, 27.0]	
Total misallocation (Input)	28.8	31.2	32.2	569.0	20.4	33.4	28.8
	[16.4, 251.2]	[15.6, 271.9]	[17.4, 766.8]	[199.2, 1205.9]	[13.4, 208.6]	[17.8, 333.7]	
Optimum (Input)	28.1	32.4	34.0	427.5	21.2	32.3	27.5
	[15.0, 248.2]	[25.6, 335.1]	[20.4, 812.2]	[171.6, 902.0]	[13.7, 203.4]	[16.0, 336.0]	
Financial frictions	4.1	9.1	4.1	4.5	5.9	4.9	7.3
	[1.6, 7.7]	[2.3, 28.9]	[-0.5, 7.8]	[-0.9, 7.2]	[1.5, 9.5]	[1.1, 9.2]	
Input frictions	5.9	7.5	7.6	0.7	3.3	7.0	4.5
	[1.0, 13.0]	[1.6, 38.6]	[0.6, 24.7]	[-0.3, 6.3]	[1.0, 27.4]	[1.4, 17.7]	
Interaction	1.3	0.5	1.1	2.6	0.3	1.0	2.9
	[0.0, 3.0]	[3.4, 41.7]	[0.4, 10.1]	[0.6, 7.5]	[0.1, 6.5]	[0.1, 5.8]	

Point estimates with 90% bootstrap confidence intervals in brackets below. Misallocation and decomposition rows report percentage gains in aggregate TFP.

5.4.3 Taxes vs. Rations

While the estimation procedure doesn't require taking a stand on whether input wedges operate as taxes or rations, this affects how households would adjust input demands in response to changes in the wedges. In particular, a household facing a downward labor ration, as in Breza et al. (2021), would not use additional credit to hire more labor. The results in Figure 4 treat all inputs as flexible, as if input frictions functioned as taxes. Figure C2 shows the decomposition of gains from reallocation if land were a fixed factor or labor were rationed from below, relative to the case where both factors are mobile yet subject to distortions. The blue (left) bars in each group reproduce the results from the baseline case of Figure 4. The green (middle) bars show the results assuming land is a fully fixed factor and the purple (right)

bars show results assuming all households with $\tau < 1$ for labor face a downward ration. The inability of land to adjust reduces the potential gains from reallocation by almost half when removing all other market failures, more so when removing only financial frictions and less so when relaxing other input frictions. However, very similar gains can be achieved even when maintaining a labor ration, as households facing a downward labor ration could use additional credit to acquire other inputs until the ration no longer binds.

6 Conclusion

In this paper, I estimate distinct distortions affecting farm households in Thailand and decompose how they each contribute to misallocation. In particular, policies to improve allocative efficiency require knowledge of the joint distribution of productivity and all distortions, as the welfare effects of interventions in a single market are ex-ante ambiguous. To identify and estimate these distortions, I develop novel methods leveraging the non-separability of households' consumption and production problems when misallocation exists. Empirically, I find relatively modest levels of misallocation in Thai agriculture: In my preferred specification, the estimated gains from optimal reallocation are 11.3%. I further decompose these gains into 4.1% from correcting financial misallocation alone, 5.9% from correcting factor misallocation alone, and the remaining 1.3% from interactions between these distortions. Factor and financial misallocation partially offset each other in equilibrium because more financially constrained farmers appear relatively subsidized in input markets, particularly for labor. This suggests that policies that seek to alleviate both input and financial distortions may be more effective than those targeted towards either set of frictions. Directly estimating financial distortions rather than inferring them from a residual also allows me to avoid attributing measurement error in inputs to misallocation. I find that not accounting for measurement error using the full model would lead to substantially larger estimates of misallocation and,in contrast to my preferred approach, suggest that removing financial frictions

alone would worsen misallocation. While the model explicitly allows for such a possibility, my preferred results show that this is not the case. This paper provides general tools that can be adapted to other settings. However, it remains agnostic to the underlying institutions behind the distortions it measures and what policies. More research is needed to understand how the effects of specific policies to address misallocative institutions in different contexts depend on the underlying distribution of distortions.

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A Full Dynamic Program

A.1 Production

Production is given by

$$Y_{it+1} = A_{it}F(q_{it}, \varphi_{it+1}) \tag{16}$$

where q_{it} is a vector of K inputs applied by i at time t, A_{it} is household TFP,, and φ_{it+1} is a shock realized at $t+1$, prior to harvesting output Y_{it+1}. As is standard, I assume that $F_k > 0, F_\varphi > 0$,and $F_{kk} < 0$ for each k. I assume that F is common across households and fixed over time, but households may have heterogeneous time-varying productivity. Note that I treat all inputs as static – in a benchmark economy with complete rental markets, households' input use at time t would not depend on their endowments or previous seasons' input choices.

I assume that $\bar{w}_{vkt}$ is the (endogenously determined) market price of each input k in village v at time t. However, households may face idiosyncratic taxes or subsidies such that they face prices $s_{ikt}\bar{w}_{vkt}$. Households may also be subject to upward or downward rations on inputs such that $\underline{q}_{ikt} \leq q_{ikt} \leq \bar{q}_{ikt}$.

While I only directly model the agricultural sector, allowing households to earn income from other sources is important to match the income diversification observed in the data. Households can invest in a portfolio of assets b_{imt} with uncertain returns r_{imt+1}, with borrowing reflected by a negative b. However, this may be subject to borrowing constraints $\bar{B}_{it}$ such that $\sum_m b_{imt} \geq \bar{B}_{it}$. b_{it} should also be thought of as capturing formal and informal insurance with state-contingent payouts. Like with inputs, frictions in the asset market can be modeled by writing returns as $r_{imt+1} \equiv \chi_{imt}\bar{r}_{vmt+1}$, where $\bar{r}_{vmt+1}$ is the (endogenously determined and possibly stochastic) average return in village v.[39] Let B_{it} denote a household's portfolio of assets and R_{it+1} be the return to that portfolio. I denote the set of primitive taxes and

[39] $\chi_{imt} = -\infty$ implies a household never purchases asset m.

rations that generate the distortions I derive below as $\mathcal{D} \equiv \{\chi, s, \underline{q}, \bar{q}, \bar{B}\}$.[40] Note that the estimation strategy I develop in section 4 does not depend on which frictions in $\mathcal{D}$ generate λ and τ. In section 5, I discuss how whether input frictions act as taxes or rations affects the decomposition and compute results both ways.

A.2 Dynamic Program

I assume households j have time-separable, von Neumann-Morgenstern preferences with discount factor δ and per-period utility function $u(c, l)$, which I assume is continuously differentiable, strictly increasing, and concave in consumption c and leisure l. At time t, they maximize

$$\mathbb{E}_t \left[\sum_{s=t}^{\infty} \delta^{s-t} u(c_{is}, l_{is}) \right]$$

subject to the following budget constraint,

$$M_{it+1} = M_{it} + Y_{it+1} - w'_{it} q_{it} - p'_t c_{it} + R_{it} B_{it} - B_{it+1} \tag{17}$$

which holds in each state of the world.

The household's value function satisfies the Bellman equation

$$V(Y, M, w, p, \varphi, R, \mathcal{D}) = \max_{c,q,B} u(c) + \delta \mathbb{E}_t V(Y', M', w', p', \varphi', R', \mathcal{D}') \tag{18}$$

subject to the budget constraint (17), borrowing constraint $\bar{B}$, and possible rations on hiring inputs in or out, $\underline{q}, \bar{q}$. Taking first-order conditions with respect to the choice variables c, q, and B:

[40] While the elements of $\mathcal{D}$ cannot be separately identified without many additional assumptions, they microfound the distortions the markets in credit, insurance and the k input markets I derive below.

$$(c) \qquad u_i(c) = \lambda p_i \tag{19}$$

$$(q) \qquad \delta \mathbb{E}\left[\frac{\partial V}{\partial Y}(Y', M', w', p', \varphi', R', \mathcal{D}')F_k(q, \varphi')\right] = \lambda w_k + \underline{\mu}_k - \bar{\mu}_k \tag{20}$$

$$(B) \qquad \delta \mathbb{E}\left[R' \frac{\partial V}{\partial M'}(Y', M', w', p', \varphi', R', \mathcal{D}')\right] + \mu^B = \lambda \tag{21}$$

where λ, μ^B, $\underline{\mu}_k$, and $\bar{\mu}_k$ are the Lagrange multipliers on the budget constraint, borrowing constraint $\bar{B}$, and rations on hiring inputs in and out, $\underline{q}, \bar{q}$, respectively. The first FOC simply states that households equate the marginal utility of expenditure on each good consumed within a period to a common Lagrange multiplier λ. The second implies that households equate the marginal utility of expenditure on each input to the expected marginal utility of its marginal product, unless an input ration binds. The third is simply the Euler equation with the possibility of binding borrowing constraints.

A.3 Input Demands and Wedges

Applying the envelope theorem to the first-order condition for q with simple substitutions yields the following expression for input demands:

$$\bar{w}_{vkt}\tau_{ikt} = \delta \mathbb{E}_t \left[F_k(q_{it}, \varphi_{it+1})\right]\Lambda_{ikt} \tag{22}$$

in which

$$\tau_{ikt} \equiv s_{ikt} + \frac{\underline{\mu}_{ikt} - \bar{\mu}_{ikt}}{\lambda_{it}\bar{w}_{vkt}} \tag{23}$$

$$\Lambda_{ikt} \equiv \frac{\mathbb{E}_t[\lambda_{it+1}]}{\lambda_{it}} + \frac{cov_t(\lambda_{it+1}, F_k(q_{it}, \varphi_{it+1}))}{\lambda_{it}\mathbb{E}_t[F_k(q_{it}, \varphi_{it+1})]} \tag{24}$$

(22) simply states that households equate the marginal utility of expenditure on input k to the discounted expected marginal utility of its marginal product. Under input frictions, the (shadow) cost of each input k differs from the common market price by τ_{ikt} as defined by

(23). Meanwhile, Λ captures how credit constraints and uninsured risk affect input demands through the two terms in (24), respectively. When credit constraints bind, (Equation 20) implies that $\lambda_{it} > \mathbb{E}_t[\lambda_{it+1}]$ since households cannot borrow against expected future earnings. Likewise, absent full insurance, consumption at $t+1$ will depend on the realization of production shocks, creating a non-zero covariance between λ_{it+1} and (stochastic) marginal products, $F_k(q_{it}, \varphi_{it+1})$. This covariance may differ across inputs for a general production function. However, it will be negative if households are prudent ($u'''(c) > 0$), input k does not reduce risk ($F_{k\varphi} \geq 0$), and agriculture is not a hedge against overall portfolio risk. In this case, both mechanisms would reduce input demands relative to the case of perfect financial markets.

Λ_{ikt} and τ_{ikt} fully characterize the distortions generated by $\mathcal{D}$ in the markets for each input k. To see this, compare (22) to the benchmark of perfect markets, in which it reduces to expected profit maximization.

$$\bar{w}_{vkt} = \delta \mathbb{E}_t[F_k(q_{it}, \varphi_{it+1})] \tag{25}$$

This is identical to (22) when $\Lambda_{ikt} = \tau_{ikt} = 1$ for all j, k, t. In this case, ratios of marginal utilities λ are constant across households and cancel out and all households equalize expected marginal products to the common price of each input ($\tau = 1$). The equalization of marginal products across households implies the allocation is efficient. Note how deviations from efficiency are completely characterized by Λ_{ikt} and τ_{ikt}, which together define the distortions in the market for each input k.

I have thus far kept the model as general as possible to illustrate how financial and input frictions create distinct wedges under very general conditions. However, estimating the model requires functional form assumptions for F and u. While I discuss functional forms for utility in section 4, I assume output is determined by the following Cobb-Douglas production function:

I assume production takes the following Cobb-Douglas form

$$Y_{it+1} = e^{\psi_{it+1}} \prod_k Q_{ikt}^{\alpha_k}. \tag{26}$$

Without loss of generality, I assume $\psi_{it+1} = a_{it} + \phi_{it+1}$, where a_{it} is anticipated productivity and ϕ_{it+1} is an unanticipated shock that cannot be predicted with time t information. I assume further a_{it} follows a first order Markov process — $a_{it} = h(a_{it-1}) + \xi_{it}$ — in which $h(\cdot)$ is a general function capturing the anticipated evolution of productivity and ξ_{it} is the unpredictable "innovation" to productivity, orthogonal to household's prior information sets by construction. I assume decreasing returns to scale with $\gamma \equiv \sum_k \alpha_k < 1$.[41]

Under the Cobb-Douglas assumption. I can rewrite (22) to obtain the demand function for each input k.

$$q_{ikt} = \frac{\delta \alpha_k}{\bar{w}_{vkt}\tau_{ikt}} \frac{\mathbb{E}_t[\lambda_{it+1} Y_{it+1}]}{\lambda_{it}} \tag{27}$$

(4) can also be expressed as

$$q_{ikt} = \frac{\delta \alpha_k}{\bar{w}_{vkt}\tau_{ikt}} \mathbb{E}_t[Y_{it+1}]\Lambda_{it} \tag{28}$$

where $\Lambda_{it} = \dfrac{\mathbb{E}_t[\lambda_{it+1}\varphi_{it+1}]}{\lambda_{it}}$ is now constant across inputs k.[42]

Meanwhile, distortions in the market for each input k enter through τ_{ikt}. In contrast, financial frictions Λ_{it} distort the *scale* of production while the *composition* of inputs is only

[41] If $\gamma \geq 1$, then the efficient allocation is degenerate with only the most productive producer producing.

[42] To see this, it is useful to write the expectation in the numerator as $\mathbb{E}_t[\lambda_{it+1}] + cov_t(\lambda_{it+1}, \varphi_{it+1})$ (since φ is mean 1 by construction). Also note that (28) can be written in closed-form by substituting the production function for Y_{it+1} and solving the system of equations implied by (4)

$$q_{ikt} = \frac{\alpha_k}{\bar{w}_{vkt}\tau_{ikt}} \left(\delta A_{it} \Lambda_{it} \prod_l \left(\frac{\alpha_l}{\bar{w}_{vlt}\tau_{ilt}} \right)^{\alpha_l} \right)^{\eta}$$

where $\eta \equiv \frac{1}{1-\gamma}$

distorted by τ. To see this, take the ratio of demands for any two inputs, k and l:

$$\frac{q_{ikt}}{q_{ilt}} = \frac{\alpha_k}{\alpha_l} \frac{\bar{w}_{vlt}}{\bar{w}_{vkt}} \frac{\tau_{ilt}}{\tau_{ikt}} \tag{29}$$

Input ratios are solely a function of technology and relative market prices, which under perfect markets are constant across households in the same village-year. Thus any dispersion in input ratios can be attributed to τ.[43] This is a feature of any homothetic production function.[44]

A.3.1 Nonhomothetic Production

While the misallocation literature typically assumes a homothetic production function with Hicks-neutral shocks, this implies that all inputs contribute proportionally to the variance as outputs. Maintaining this assumption not only increases tractability, but allows me to directly compare my results to others in the literature, showing how modeling the consumption side produces drastically different conclusion, holding the model fixed. However, a stark implication of homotheticity is that households facing the same input prices would use the same input mix and financial distortions would only affect the scale of production. To relax this assumption, I assume production takes the following generalized Cobb-Douglas form following Just and Pope (1978) and Just and Pope (1979).

$$Y_{it+1} = A_{it} \prod_k^K q_{ikt}^{\alpha_k} + \phi_{it+1} B_t \prod_k^K q_{ikt}^{\beta_k} \tag{30}$$

where Y_{t+1} is output realized the period following production, q_{kt} is the quantity of input k at time t, A is TFP, and ϕ_{t+1} is a mean-zero shock realized before harvest and consumption at $t+1$. I assume that expected returns to scale $\gamma \equiv \sum_k \alpha_k < 1$ to ensure the socially optimal allocation is nondegenerate. The main difference between this and the workhorse

[43] Note that $s, \underline{q}, \bar{q}, \bar{B}$, and $\{\chi\}$ are the primitives that determine the distortions τ and Λ.

[44] Note that under CES production, the ratio of τs on the right-hand side of (29) is raised to the elasticity of substitution σ.

Cobb-Douglas specification is that the variance of output now depends on input composition. Inputs are differentially risky if $\alpha \not\propto \beta$. In particular, α_k can be thought of as the elasticity of the expectation of output with respect to input k, while β_k is the elasticity of the *standard deviation* of output with respect to input k.[45]

$$q_{ikt} = \frac{\delta\left(\alpha_k \mathbb{E}_t[Y_{it+1}]\mathbb{E}_t[\lambda_{it+1}] + \beta_k cov_t(\lambda_{it+1}, Y_{it+1})\right)}{\lambda_{it}\bar{w}_{kvt}\tau_{ikt}} \tag{31}$$

Note how when $\alpha = \beta$ this reduces to (4).

The only difference is that (31) assigns different coefficients to the expected and stochastic components of $\mathbb{E}_t[\lambda_{jt+1}Y_{jt+1}]$. Inputs with higher β contribute more to the variability of output, causing their demand to be disproportionately affected by imperfect insurance. In contrast, the separability of the shocks in the standard Cobb-Douglas means that the same Λ_{it} applies to demand for each input.[46]

The ratio of λs in the first term can be thought of as the wedge created by the inability to intertemporally smooth consumption and is constant across inputs. For example, if a household faces a binding borrowing constraint, then $\mathbb{E}_t[\lambda_{t+1}]$ would generally be lower than λ_t. The second term captures how uninsured risk affects demand. Again, one would expect the covariance term to be negative,[47] but this is amplified by how risky a given input is.

However, this no longer allows the straightforward identification of τ from (29), requiring an alternative set of identification assumptions, which I discuss in Appendix B. I also show results from this more general specification and the results are broadly similar to those under the standard Cobb-Douglas. Definition 1 formalizes an equilibrium in either case.

Definition 1. *A decentralized equilibrium is defined by a set of prices* $\{\bar{w}_{vkt}, p_{it}, R_v t\}$*, an input allocation* $\{q_{ikt}\}$*, and a consumption allocation* $\{c_{it}\}$ *such that*

[45] I prefer this specification to that recently introduced by Bohr et al. (2023), since this functional form allows for a first order effect of uninsured risk on input demand as shown below. Note that this functional form nests the workhorse Cobb-Douglas specification $Y_{t+1} = A_t e^{\phi_{t+1}} \prod_{k=1}^{K} q_{kt}^{\alpha_k}$ if $\alpha = \beta$ and $B = A/\mathbb{E}[e^{\phi}]$.

[46] This is true for any homothetic production function.

[47] unless $u'''(c) \leq 0$ or returns from agriculture are sufficiently negatively correlated with those from other investments

1. *Households optimize following (19)-(21)*

2. *Input demands* q_{ikt} *equal* $\omega_{ikt}\bar{Q}_{vkt}$*, where* ω_{ikt} *is given by (5) and* $\sum_{i=1}^{N_v}\omega_{ikt} = 1$ *for each* v

3. Λ_{it} *and* τ_{ikt} *are defined as in (23) and* (28)

given a set of initial asset holdings M_{it} *and primitive distortions* $\mathcal{D}$*.*

B Generalized Cobb-Douglas Production

B.1 Estimating α and β

I now describe how each of the key components of the model λ, τ, α,β A, and Λ are estimated. Since the estimation of marginal utilities, λ, doesn't depend on the production function, I follow the same procedure as in Section 4.1.2. I then estimate (31) from a subsample of households' input demands. Once I recover α and β, I then recover τ, Λ and A for the full sample of households.

I use GMM to estimate α and β from the system of input demand equations defined by (31). In this approach, inputs are analogous to assets in a CAPM model and λ_t/λ_{t+1} is analogous to the portfolio's return (Hansen & Singleton, 1982). Under rational expectations, this yields a straightforward approach to estimation.

Let $x_{ikt} \equiv \bar{w}_{kvt}\tau_{kt}q_{ikt}$. x_{ikt} can be interpreted as household j's "shadow" expenditure on input k at time t. This can either represent actual expenditure under possibly household-specific prices or as the cost of input k such that the household would choose q_{ikt} under perfect markets. Let $\mathcal{I}_{it}$ be a vector of variables in household j's information set at time t. Rearranging constrained-optimal input demands (31) and making the dependence on households' time t information sets explicit yields the moment condition.

$$\alpha_k\mathbb{E}[\lambda_{i,t+1}|\mathcal{I}_{it}]\mathbb{E}[Y_{i,t+1}|\mathcal{I}_{it}] + \beta_k cov(\lambda_{i,t+1,}Y_{i,t+1}|\mathcal{I}_{it}) - \lambda_{it}x_{ikt} = 0 \tag{32}$$

where $cov_t(\lambda_{i,t+1}Y_{i,t+1}) = \mathbb{E}_t[\lambda_{i,t+1}Y_{i,t+1} - \mathbb{E}_t[\lambda_{i,t+1}]\mathbb{E}_t[Y_{i,t+1}]]$ can be thought of as a measure of how households expect their utility at harvest to depend on the realizations of production shocks, conditional on their time t information. Estimation requires mapping the subjective expectations $\mathbb{E}[\lambda_{it+1}|\mathcal{I}_{jt}], \mathbb{E}[Y_{it+1}|\mathcal{I}_{jt}]$, and $\mathbb{E}[\lambda_{jt+1}Y_{jt+1}|\mathcal{I}_{jt}]$ to data. The nested case of $\alpha = \beta$ in Section 4 doesn't require distinguishing between $\mathbb{E}[\lambda_{it+1}|\mathcal{I}_{jt}]\mathbb{E}[Y_{it+1}|\mathcal{I}_{jt}]$ and $E[\lambda_{jt+1}Y_{jt+1}|\mathcal{I}_{jt}]$, which allows me to substitute realized $\lambda_{jt+1}Y_{jt+1}$ for $\mathbb{E}[\lambda_{jt+1}Y_{jt+1}|\mathcal{I}_{jt}]$ under rational expectations. More formally, Differences between the expected and realized products of output and marginal utilities can be expressed as:

$$\lambda_{i,t+1}Y_{t+1} - \mathbb{E}[\lambda_{i,t+1}Y_{t+1}|\mathcal{I}_{jt}] = \zeta_{i,t+1} \tag{33}$$

I assume households are fully forward-looking and have rational expectations over future shocks. In this case $\mathbb{E}_t[\zeta_{i,t+1}|\mathcal{I}_{jt}] = 0$, as ζ is simply prediction error that arises from the realization of shocks after households' optimal decisions are made in time t. The challenge is that $\mathbb{E}_t[\zeta_{i,t+1}|\mathcal{I}_{jt}]$ is the household's *subjective* expectation as of time t, conditional on its information set $\mathcal{I}_{jt}$ but prior to the realization of shocks, and is unobserved. I do observe $\lambda_{jt+1}Y_{jt+1}$ for N households in T years. In a given year, the *population* the mean of realized ζ_{it+1}, which I denote as $\mathbb{E}[\zeta_{t+1}]$, may be nonzero if there are aggregate shocks that affect all households within a period.

This motivates a set of instruments

$$z_{it} \equiv \{\lambda_{i,t-1}, \lambda_{i,t-2}, \ldots\}$$

that are plausibly orthogonal to prediction error. The logic of this is that past consumption is correlated with future consumption, making z_{it} relevant.[48] However, whether $\mathbb{E}[\zeta_{it+1}z_t] = 0$ depends on whether their covariance is stationary. Within a given year, realizations of

[48]Importantly, these instruments are functions of consumption that takes place after the previous season's shocks are realized, meaning that they don't reflect uncertainty from previous seasons.

shocks are likely to differentially affect households with different levels of wealth, and thus z_t. Intuitively, poorer households may be more risk-averse (under prudence) and less insured, and thus their marginal utilities will be more sensitive to the realizations of shocks. The stationarity of $\mathbb{E}[\zeta_{it+1} z_t] = 0$ ensures that the effects of this greater sensitivity of poorer households averages out to zero in the panel.[49]

This assumption permits the substitution of (subjective) conditional expectations $\mathbb{E}[\lambda_{jt+1} Y_{jt+1} | \mathcal{I}_{jt}]$ with realizations. However, this substitution can't be used for both $\mathbb{E}[\lambda_{it+1} | \mathcal{I}_{jt}] \mathbb{E}[Y_{it+1} | \mathcal{I}_{jt}]$ and $\mathbb{E}[\lambda_{jt+1} Y_{jt+1} | \mathcal{I}_{jt}]$ Therefore, separately identifying α and β requires taking a stand on what shocks the household does and does not anticipate at time t.

One approach would be projecting realizations of λ_{it+1} and Y_{it+1} on to functions of $\mathcal{I}_{jt}$, say $l(\mathcal{I}_{jt})$ and $y(\mathcal{I}_{jt})$, and using the predicted values, $\hat{l}(\mathcal{I}_{jt})$ and $\hat{y}(\mathcal{I}_{jt})$,to substitute for $\mathbb{E}[\lambda_{it+1} | \mathcal{I}_{jt}]$ and $\mathbb{E}[Y_{it+1} | \mathcal{I}_{jt}]$, respectively. In this case

$$
\begin{aligned}
\lambda_{it+1} &= \mathbb{E}[\lambda_{it+1} | \mathcal{I}_{jt}] + \pi^L_{it+1} \\
Y_{it+1} &= \mathbb{E}[Y_{it+1} | \mathcal{I}_{jt}] + \pi^Y_{it+1} \\
\lambda_{it+1} &= \hat{l}(\mathcal{I}_{jt}) = +\upsilon^L_{it+1} \\
Y_{it+1} &= \hat{y}(\mathcal{I}_{jt}) + \upsilon^Y_{it+1}
\end{aligned}
\tag{34}
$$

The household's prediction errors π are mean zero by rational expectations and the estimation errors υ are mean 0 by construction. This means that the difference these two errors $\psi^Y_{it} \equiv \pi^L_{it} - \upsilon^L_{it}$ and $\psi^Y_{it} \equiv \pi^Y_{it} - \upsilon^Y_{it}$ are each mean zero by linearity of expectations. However substituting the *product* of subjective $\mathbb{E}[\lambda_{it+1} | \mathcal{I}_{jt}] \mathbb{E}[Y_{it+1} | \mathcal{I}_{jt}]$ for realizations implies:

[49]This would be violated if households' mispredictions, conditional on time t information, were correlated with income prior to time t.[50] However, I show with Monte Carlo simulations that the estimator I derive below performs well with small T, even when there is a correlation between baseline wealth and the realizations of aggregate shocks in the finite sample.

$$\mathbb{E}\left[\left(\alpha_k(\hat{l}(\mathcal{I}_{jt})+\upsilon^L_{it+1})(\hat{y}(I)+\upsilon^Y_{it+1})+\beta_k(\lambda_{jt+1}Y_{jt+1}-(\hat{l}(\mathcal{I}_{jt})+\upsilon^L_{it+1})(\hat{y}(I)+\upsilon^Y_{it+1}))-\lambda_{it}x_{ikt}\right)\otimes h(\mathcal{I}_{jt})\right]$$
$$=(\alpha_k-\beta_k)(\psi^L_{it+1}\hat{y}(\mathcal{I}_{jt})+\psi^Y_{it+1}\hat{l}(\mathcal{I}_{jt})+\psi^L_{it+1}\psi^Y_{it+1})\otimes h(\mathcal{I}_{jt})=0 \tag{35}$$

Assuming $\hat{l}(\mathcal{I}_{jt})$ and $\hat{y}(\mathcal{I}_{jt})$ provide accurate predictions of the true subjective expectations, $\mathbb{E}[\lambda_{it+1}|\mathcal{I}_{jt}]$ and $\mathbb{E}[Y_{it+1}|\mathcal{I}_{jt}]$, they will differ by estimation error υ_l and υ_y, respectively. Both of these are mean 0 and orthogonal to $\mathcal{I}_{jt}$, by construction. However, their product is not necessarily mean 0. Estimation errors are likely to be correlated absent full insurance or quadratic utility, negatively if $u'''(c)>0$ and production is not used as a hedge against portfolio risk: any anticipated productivity shock not captured by functions of the observed elements of $\mathcal{I}_{jt}$ is likely to have opposite effects on $\mathbb{E}[\lambda_{it+1}|\mathcal{I}_{jt}]$ and $\mathbb{E}[Y_{it+1}|\mathcal{I}_{jt}]$.

I assume households are fully forward-looking and have rational expectations. Therefore, by virtue of optimization, any differences between realized shocks and households' conditional expectations as of time t are mean-zero prediction errors Denote households' conditional expectations of $\lambda_{i,t+1}$ and $Y_{i,t+1}$, respectively, as $\bar{\lambda}_{i,t+1}$ and $\bar{Y}_{i,t+1}$. This implies that households' subjective expectations of random variables, on average, equal their realizations, meaning that we can substitute the conditional expectations in (32) with their unconditional expectations (Hansen & Singleton, 1982).

Letting $\mathbb{E}$ denote the unconditional expectation operator, this implies that estimating

$$\mathbb{E}\left[\left(\alpha_k\hat{l}(\mathcal{I}_{jt})\hat{y}(I)+\beta_k(\lambda_{jt+1}Y_{jt+1}-\hat{l}(\mathcal{I}_{jt})\hat{y}(I))-\lambda_{it}x_{ikt}\right)\otimes h(\mathcal{I}_{jt})\right]$$
$$=\mathbb{E}[(\alpha_k(\mathbb{E}[\lambda_{it+1}|\mathcal{I}_{jt}]\mathbb{E}[Y_{it+1}|\mathcal{I}_{jt}]+\upsilon_{it})+\beta_k(\mathbb{E}[\lambda_{jt+1}Y_{jt+1}|\mathcal{I}_{jt}]-\mathbb{E}[\lambda_{it+1}|\mathcal{I}_{jt}]\mathbb{E}[Y_{it+1}|\mathcal{I}_{jt}]-\upsilon_{it}+\zeta_{it+1})$$
$$-\lambda_{it}x_{ikt})\otimes h(\mathcal{I}_{jt})]$$
$$=((\alpha_k-\beta_k)\upsilon_{it}+\zeta_{xi})\otimes h(\mathcal{I}_{jt})=0$$

where $\upsilon_{it} \equiv \upsilon_{it}^{l}\hat{y}(\mathcal{I}_{jt}) + \upsilon_{it}^{y}\hat{l}(\mathcal{I}_{jt}) + \upsilon_{it}^{l}\upsilon_{it}^{y}$ collects errors from the auxiliary regressions and h is a measurable function of observed elements of $\mathcal{I}_{it}$ and $\otimes$ is the Kronecker product. Intuitively, the difference between households' subjective prediction errors and the econometrician's estimation errors needs to average out to zero in the sample.

Taking sample averages:

$$g_{NT}(\alpha, \beta) \equiv \frac{1}{NT}\left(\sum_{i=1}^{N}\sum_{t=1}^{T}(\alpha_k - \beta_k)\hat{l}(\mathcal{I}_{jt})\hat{y}(\mathcal{I}_{jt}) + \beta_k(\lambda_{i,t+1}Y_{i,t+1}) - \lambda_{it}x_{ikt}\right) \otimes h(\mathcal{I}_{it}), \quad (36)$$

which converges to 0 with large NT under similiar conditions as in the main analysis. Thus, the GMM estimate of $\theta = \{\alpha, \beta\}$ is

$$\arg\min_{a,b} J(a, b) \equiv g_{NT}(a, b)' W g_{NT}(a, b) \quad (37)$$

where W is the standard optimal weighting matrix.

A second challenge is separately identifying τ, since households facing common technology and prices will no longer necessarily have the same input ratios. To make progress, I draw on empirical IO methods to estimate product-level production functions with unobserved input prices. In the case of De Loecker et al. (2016), they observe single- and multi-product firms producing the same goods but only observe inputs at the firm level. Their solution is to estimate the production function restricting the sample to single-product firms, and then apply a selection correction to control for unobservable differences between these two types of firms.

The problem in my case is that τ is not necessarily observed. Depending on the nature of input distortions, τ may correspond to the difference between the market price of an input and the price actually paid by a household that purchases this input, or it may be a shadow price that a household faces when rationed. I observe both input expenditures and quantities in the data. I assume that when households hire labor or rent land, any distortion is reflected in the observed price they pay. In this case, τ_{ikt} is included in the x_{ikt} I observe, which is the

appropriate variable for (36). Thus I restrict the sample to transacted inputs when estimating α and β, which I then use to recover τs for the households that do not transact these inputs. Note that I do not have to make such assumptions about the nature of τs when production is homothetic, as I can estimate these directly from factor ratios.[51]

C Additional Tables and Figures

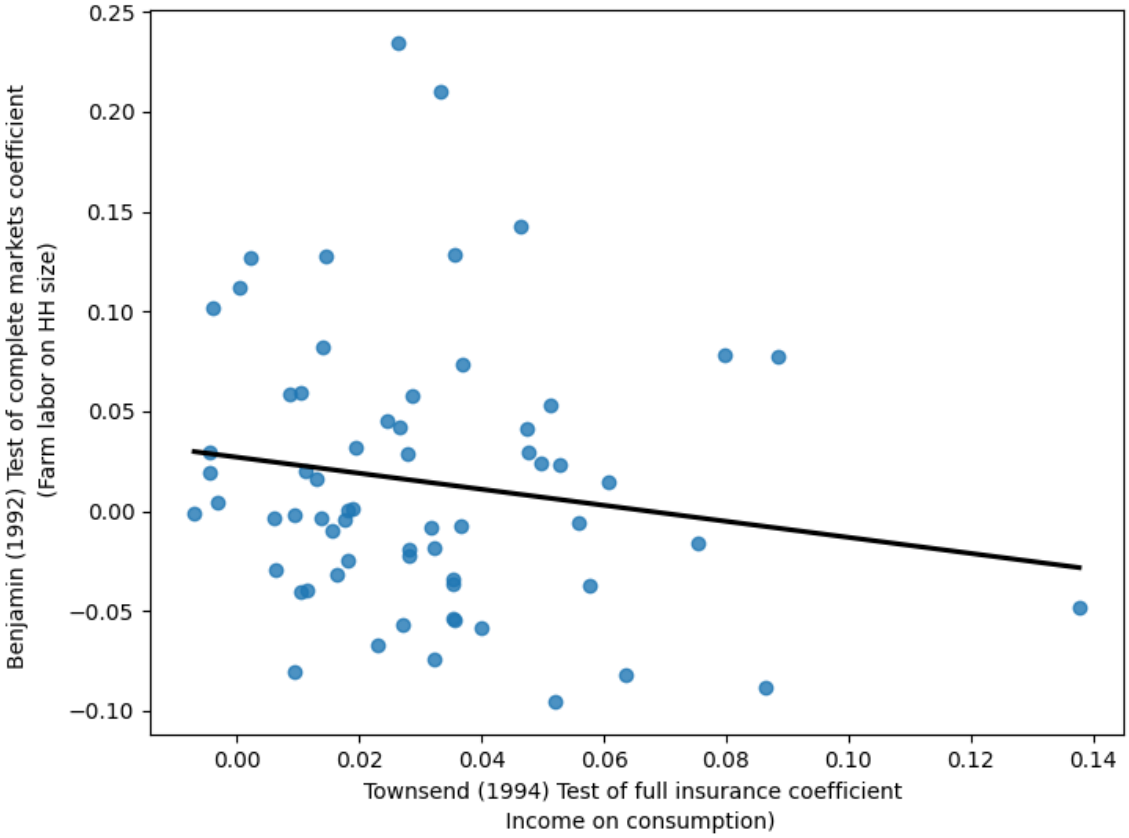
[51] This approach relies on some strong assumptions — namely that there is no selection into hiring inputs, that transacted inputs have the same returns as those owned by the household, and that households who purchase positive amounts of inputs do not come up against a ration. To provide support for the first assumption, I can apply the control function approach in De Loecker et al. (2016). I can also restrict the sample to households that use their own inputs in some seasons and purchase inputs in others. To address the second, I observe individual laborer and plot identifiers and can test whether their observed productivity differs when they are used by their respective households or hired. The third assumption is more difficult to test, but I can attempt to restrict the sample to households that appear less likely to face a binding ration.

Table C1: Diagnostic Tests for Market Failures

	log Consumption Val.	log Labor Hrs.	
	(1)	(2)	(3)
log Income	0.0547***		
	(0.0037)		
HH Size		0.0211*	
		(0.0112)	
Male adults			0.0257
			(0.0258)
Female Adults			0.0269
			(0.0253)
Male children			0.0121
			(0.0217)
Female Children			0.0165
			(0.0210)
Household FE	Yes		
Village-month FE	Yes		
Village-year FE		Yes	Yes
F-stat			11.61**
p-val			0.0205
Observations	83,384	5,689	5,689

This table presents the results for two of the canonical tests of incomplete markets in the literature. Column (1) shows the results of a regression of (log) consumption on income with household and village-month fixed effects as in Townsend (1994). The full monthly sample of households (agricultural and non-agricultural) and monthly measures of total income and consumption are used. Column (2) shows the results of the Benjamin (1992) test of separability, which regresses (log) household labor hours on household characteristics, controlling for farm size. For simplicity, household size is the only measured included. Data from the full sample of producers aggregated to the household-year level are used.

Figure C1: Comparison of Test Coefficients Across Villages



This figure contains a scatter plot of the coefficients from the Townsend (1994) and Benjamin (1992) tests, run separately for each village in each 48-month block of the full panel. For the Townsend coefficients on the x-axis, the full monthly sample of households (agricultural and non-agricultural) and monthly measures of total income and consumption are used. Data from the full sample of producers aggregated to the household-year level are used to estimate the Benjamin coefficients on the y-axis.

Table C2: Summary statistics for agricultural households by township

	Full Sample	Chachoengsao	Lopburi	Buriram	Sisaket
HH Size	5.66	5.91	5.58	5.18	6.01
	(2.38)	(3.08)	(2.3)	(2.05)	(2.35)
Age Head	56.23	61.08	52.02	53.60	59.92
	(13.3)	(14.19)	(12.75)	(12.49)	(12.47)
Sex Head	0.82	0.79	0.86	0.85	0.79
	(0.38)	(0.41)	(0.35)	(0.35)	(0.41)
Head Primary Educ	0.87	0.93	0.71	0.95	0.93
	(0.33)	(0.26)	(0.45)	(0.23)	(0.26)
Head Secondary Educ	0.09	0.06	0.07	0.11	0.09
	(0.28)	(0.23)	(0.26)	(0.32)	(0.28)
Formal Loan	0.32	0.10	0.43	0.35	0.31
	(0.51)	(0.31)	(0.57)	(0.49)	(0.52)
Any Loan	0.70	0.49	0.71	0.75	0.78
	(0.46)	(0.5)	(0.46)	(0.43)	(0.41)
Years in Ag	11.07	7.18	10.26	11.11	14.43
	(5.62)	(6.28)	(5.22)	(5.15)	(3.74)
Ag Income (000s)	73.94	94.94	28.18	145.48	35.68
	(144.65)	(169.81)	(52.89)	(212.33)	(33.66)
Wage Income (000s)	38.82	85.43	22.64	44.58	19.28
	(72.62)	(99.89)	(43.58)	(74.84)	(57.2)
Other Ent. Income (000s)	120.77	251.85	77.43	171.79	28.03
	(340.98)	(521.68)	(300.66)	(354.09)	(82.94)
Has Formal Bank Acct	0.13	0.09	0.12	0.15	0.13
	(0.33)	(0.29)	(0.32)	(0.36)	(0.34)
Savings Bal. (000s)	35.40	70.84	25.26	43.93	14.07
	(107.74)	(167.77)	(54.02)	(132.44)	(48.89)
Loan Amount (000s)	49.52	59.78	37.30	72.97	31.81
	(91.62)	(127.91)	(65.39)	(111.24)	(48.51)
Gifts Made (000s)	4.98	10.09	3.67	5.54	2.42
	(6.4)	(8.69)	(4.54)	(6.48)	(3.57)
Gifts Received (000s)	10.87	18.40	7.69	14.19	5.87
	(51.12)	(72.77)	(40.77)	(61.15)	(25.02)
Consumption (000s)	80.42	120.41	70.26	96.71	48.15
	(65.84)	(82.34)	(53.04)	(70.45)	(34.85)
N Households	621	109	181	168	163

This table shows summary statistics for agricultural households by township. The table displays means and standard deviations for each variable averaged across household-year observations.

Table C3: Summary statistics for agricultural production by township

	Full Sample	Chachoengsao	Buriram	Lopburi	Sisaket
Rice	0.68	0.75	0.95	0.03	0.96
	(0.47)	(0.44)	(0.23)	(0.18)	(0.2)
Maize	0.14	0.01	0.01	0.49	0.00
	(0.35)	(0.11)	(0.1)	(0.5)	(0.04)
Farm size (ha)	4.78	5.76	2.35	9.56	2.55
	(7.93)	(5.71)	(1.76)	(13.2)	(1.84)
# plots	3.29	3.02	2.16	4.69	3.14
	(2.83)	(2.45)	(1.32)	(4.07)	(2.03)
Any plot rented	0.24	0.34	0.27	0.37	0.08
	(0.43)	(0.48)	(0.44)	(0.48)	(0.27)
Any equip. rented	0.67	0.46	0.68	0.74	0.68
	(0.47)	(0.5)	(0.47)	(0.44)	(0.47)
Any labor hired	0.69	0.65	0.79	0.88	0.49
	(0.46)	(0.48)	(0.41)	(0.33)	(0.5)
% labor hired	0.32	0.18	0.33	0.57	0.15
	(0.33)	(0.21)	(0.29)	(0.35)	(0.23)
Any fert.	0.88	0.87	0.92	0.82	0.91
	(0.32)	(0.34)	(0.27)	(0.38)	(0.28)
Any equip.	0.96	0.83	0.98	0.95	1.00
	(0.19)	(0.37)	(0.14)	(0.22)	(0.06)
Output (000s)	129.42	236.24	52.76	265.36	45.58
	(304.86)	(279.67)	(152.26)	(495.31)	(40.6)
Input exp. (000s)	87.80	115.87	43.11	178.17	41.58
	(153.66)	(115.81)	(49.39)	(253.88)	(27.76)
Land share	0.33	0.20	0.39	0.28	0.38
	(0.16)	(0.13)	(0.14)	(0.15)	(0.13)
Plant labor share	0.06	0.09	0.06	0.04	0.07
	(0.07)	(0.08)	(0.07)	(0.05)	(0.06)
Weed labor share	0.04	0.12	0.02	0.05	0.02
	(0.08)	(0.15)	(0.06)	(0.08)	(0.03)
Harv. labor share	0.16	0.11	0.18	0.18	0.15
	(0.13)	(0.2)	(0.11)	(0.14)	(0.1)
Equip. share	0.21	0.25	0.18	0.24	0.18
	(0.13)	(0.16)	(0.09)	(0.15)	(0.11)
Fert. share	0.10	0.14	0.09	0.08	0.10
	(0.08)	(0.1)	(0.07)	(0.09)	(0.06)
Seed share	0.11	0.09	0.08	0.14	0.10
	(0.08)	(0.07)	(0.05)	(0.11)	(0.05)
Profit share	0.17	0.79	0.03	0.18	0.06
	(0.67)	(0.8)	(0.51)	(0.78)	(0.49)
N Households	621	109	181	168	163

This table shows summary statistics for agricultural households by township. The table displays means and standard deviations for each variable averaged across household-year observations.

D More on CFE Demands

I provide additional details on the CFE demand system of Ligon (2020) used for the main results. CFE demands satisfy the condition that $\log p_i c_i = a_i(p) + b_i(z) - \beta_i \log \lambda$, where

Table C4: Coefficients of variation in factor and output prices by township

	Chachoengsao	Lopburi	Srisaket
Land rent (per rai)	0.522000	0.439000	0.476000
Wage (hourly)	0.718000	0.565000	0.992000
Planting wage (hourly)	0.682000	0.472000	0.854000
Weeding wage (hourly)	0.590000	0.531000	0.583000
Harvest wage (hourly)	0.615000	0.548000	0.921000
Price of rice seed (per kg)	0.266000	0.207000	0.110000
Price of chem. fert. (per kg)	0.178000	0.141000	0.095000
Power tiller rental (per rai)	0.417000	0.472000	0.721000
Large tractor rental (per rai)	0.296000	0.419000	0.288000
Output price of rice (per kg)	0.094000	0.115000	0.085000

This table shows the coefficients of variation of input and output prices within each township averaged across years. The top panel shows the inputs that I assume are distorted, while the bottom panel shows those that I assume are freely traded. The coefficients of variation are computed at the township-year level after trimming outlier per-unit plot-level expenditures at the upper and lower 2.5% tails and restricting the sample to inputs/outputs with at least 20 observations within a township-year. The three townships shown are those that nearly universally produce rice. The data do not contain the number of days that tractors or power tillers are used — therefore the unit prices I compute are the total expenditure for each type of machinery at the plot level divided by the plot area. Therefore, much of the price dispersion depicted is likely to result from number of days used, machine sizes, or measurement error. Since a more diverse range of crops is grown in Buriram, there is additional heterogeneity due to varieties of seed and fertilizers used for different crops (which I observe). When accounting for this heterogeneity, similar patterns of high price dispersion in land and labor but low price dispersion for traded inputs and outputs emerge.

Table C5: GMM results proxying for anticipated productivity shocks

	α	Const	Inp Intensity	M. Adults	F. Adults	M. Kids	F. Kids
Equip.	0.207***	0.013	-0.260	-0.011	0.142*	0.147*	0.132*
	(0.019)	(0.314)	(0.173)	(0.064)	(0.068)	(0.076)	(0.067)
Fert.	0.039***	-0.285	0.222	-0.083	0.154*	0.171	0.154
	(0.011)	(0.507)	(0.148)	(0.071)	(0.073)	(0.099)	(0.090)
Harv. Labor	0.139***	-0.379	0.266	0.223	0.336	-0.073	-0.057
	(0.016)	(0.636)	(1.073)	(0.205)	(0.268)	(0.085)	(0.060)
Land	0.258***	0.123	-0.610***	-0.008	0.124**	0.094	0.092
	(0.023)	(0.280)	(0.122)	(0.049)	(0.053)	(0.053)	(0.053)
Plant. Labor	0.071***	-0.203	-0.411***	-0.007	0.120*	0.107	0.098
	(0.008)	(0.436)	(0.083)	(0.056)	(0.065)	(0.064)	(0.064)
Seed	0.104***	0.168	-0.359**	-0.014	0.129*	0.128*	0.090
	(0.011)	(0.468)	(0.112)	(0.050)	(0.062)	(0.059)	(0.058)
Weed. Labor	0.032***	-0.447	-0.222**	-0.093	0.177**	0.062	0.225*
	(0.007)	(0.332)	(0.077)	(0.068)	(0.075)	(0.109)	(0.106)
γ	0.850	ζ	2.444				
γ s.e.	(0.045)	ζ s.e.	(0.169)				
J	54.915						
p-val	0.072						

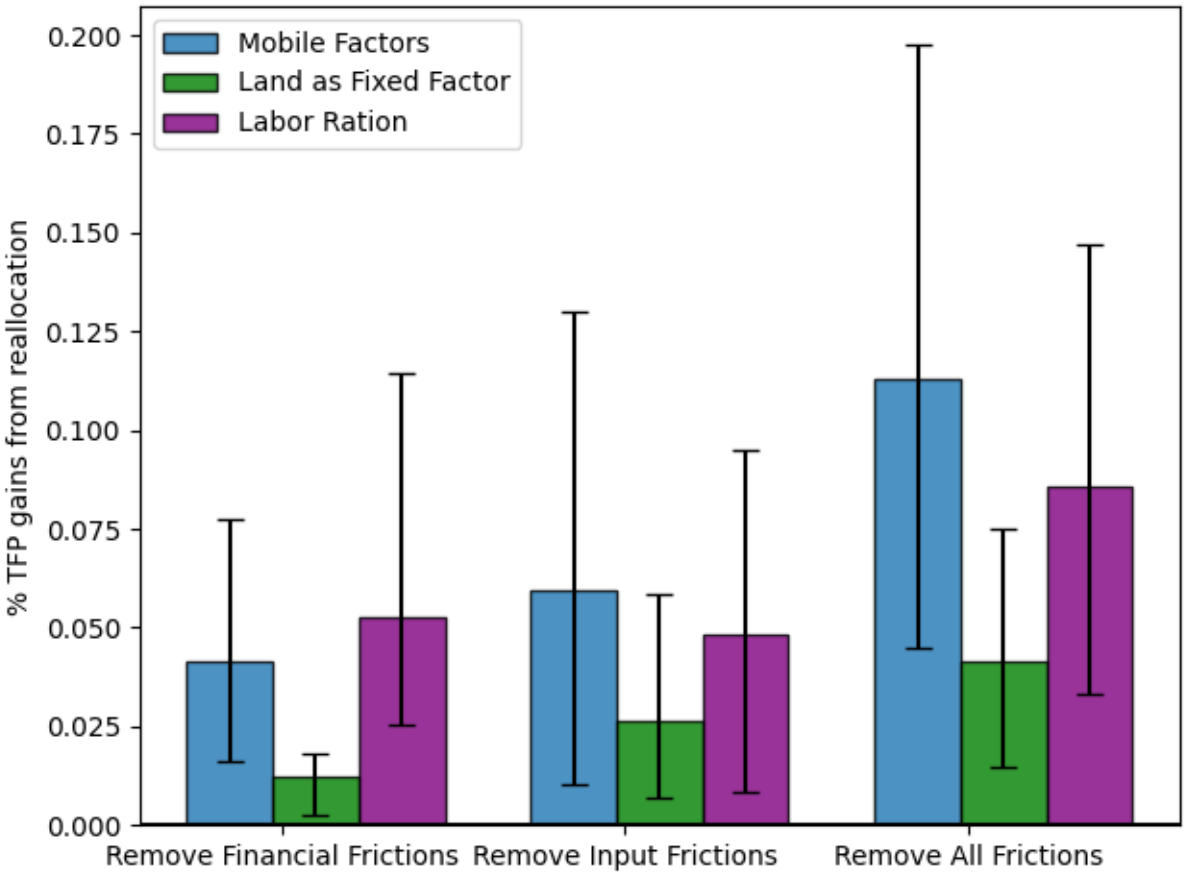
This table reports GMM estimates from the specification that includes post-Lasso-selected proxies for anticipated productivity shocks. The selected proxies and their coefficients are reported in Table C6. All other moment conditions and estimation details are identical to the baseline specification in Table 1.

Table C6: Candidate and selected features for proxying for shocks to anticipated productivity

	Selected	Coef	s.e.
1 mo. lag of λ	✓	-0.017	(0.016)
2 mo. lag of λ			
3 mo. lag of λ	✓	-0.001	(0.017)
4 mo. lag of λ			
5 mo. lag of λ			
Ag. and biz. assets	✓	0.002	(0.003)
Cash on hand	✓	0.010	(0.006)
Current λ			
Deposits in financial institutions	✓	-0.000	(0.003)
Dummy for any non-farm revenue			
Dummy for savings from non-operating income			
Female adults	✓	0.001	(0.012)
Female kids			
Gifts and transfers made			
Gifts and transfers received			
Household assets	✓	-0.000	(0.006)
Inventories	✓	0.001	(0.006)
Land and land improvements	✓	0.002	(0.003)
Liability to asset ratio			
Male adults	✓	-0.002	(0.011)
Male kids			
Non-farm costs	✓	-0.001	(0.004)
Non-farm revenue	✓	0.003	(0.004)
Other operating loss	✓	-0.001	(0.003)
Savings from non-operating income	✓	-0.001	(0.002)

This table shows the candidate features for proxying anticipated productivity shocks using elements of households' $t-1$ information sets, whether it was selected by the preliminary multiple-post Lasso step, and the coefficient and standard error of π from the GMM estimation if it was selected. All variables are from the prior production season and are logged, with the exception of household composition counts and dummies. None of the $\hat{\pi}$s are statistically significant.

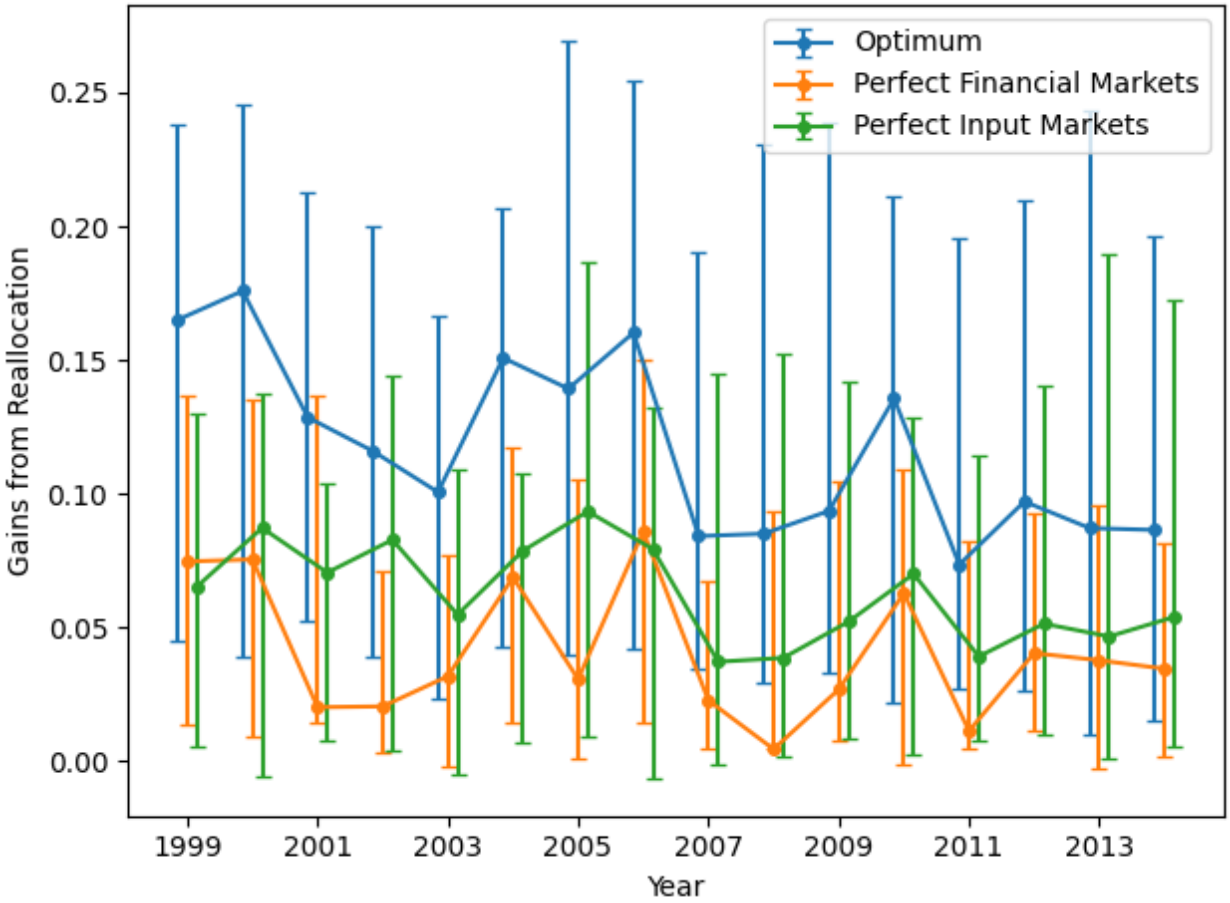
Figure C2: Counterfactual gains from reallocation under input rations



The figure shows the gains from reallocation under the main counterfactuals depending on which factors are mobile within townships. The blue (left) bars reproduce the baseline scenario, in which all factors are mobile and can be reallocated. The green (middle) bars show results holding land fixed at observed levels in all three scenarios, even when relaxing other input frictions. The purple (right) bars show results assuming households with $\tau < 1$ for each labor input face a binding downard ration. Results are computed using CFE demands, only rice crops, and aggregating to the farm level. 90% confidence intervals from 200 bootstrap replications are plotted.

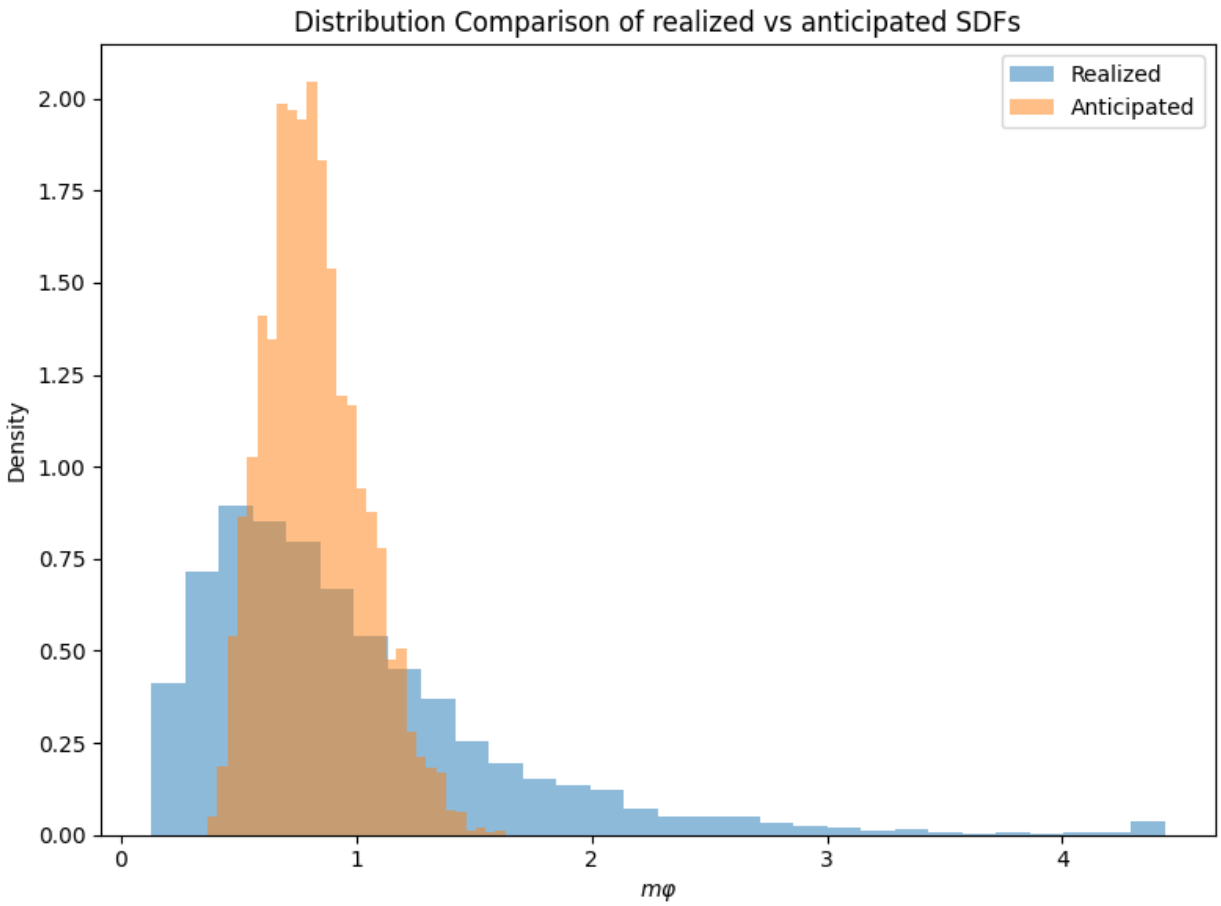
expenditures on good i depend on functions of the price vector p and household characteristics z and are log-linear in λ. β_i is the eponymous constant elasticity, which imposes that the elasticity of expenditure on good i with respect to the marginal utility of expenditure (as opposed to total expenditure) is a constant. This allows for highly non-linear Engel curves and an unrestricted rank of the demand system. Ligon (2020) shows that CFE is the only globally regular demand system in which identical households with different budgets' demands for goods differ only through a common aggregator. The paper also derives an estimator for the MUE that uses disaggregated consumption data. The key assumption for estimation is

Figure C3: Main results over time



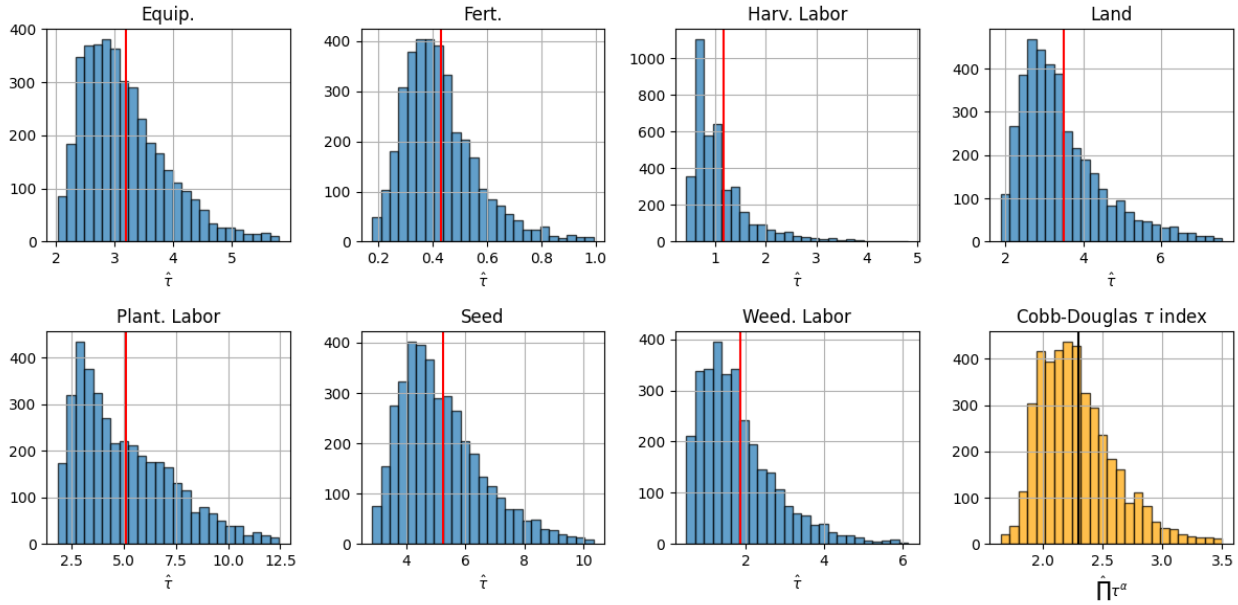
This figure plots the main results in Figure 6 disaggregated by year. Results are computed using CFE demands, restricting the sample to rice plots, and aggregating to the farm level. The measure of misallocation is the difference between aggregate TFP under a given allocation and the efficient one, expressed as a percent of modeled TFP. 90% confidence intervals from 200 bootstrap replications are plotted.

Figure C4: Distribution of realized and predicted $m\varphi$



This figure plots the distributions of the raw and predicted values of $\frac{\lambda_{it+1}}{\lambda_{it}}\varphi$ following the random forest.

Figure C5: Histogram of estimated τ_k and index of τs



This figure plots histograms of the estimated $\hat{\tau}_k$ for each input as well as the Cobb-Douglas price index of τs. The 1% tails are omitted for clarity.

Table C7: GMM results with labor as a single input

	α	Const	Inp Intensity	M. Adults	F. Adults	M. Kids	F. Kids
Equip.	0.201***	-0.412*	-0.484***	-0.027	0.111**	0.049	0.066*
	(0.026)	(0.212)	(0.067)	(0.033)	(0.039)	(0.037)	(0.036)
Fert.	0.054***	-0.651	-0.016	-0.071*	0.140**	0.044	0.078
	(0.016)	(0.418)	(0.058)	(0.037)	(0.043)	(0.045)	(0.049)
Labor	0.190***	-0.232	-0.558***	-0.029	0.122***	0.018	0.053
	(0.020)	(0.222)	(0.053)	(0.030)	(0.036)	(0.031)	(0.033)
Land	0.272***	0.235	-0.405***	-0.012	0.097**	0.015	0.043
	(0.031)	(0.193)	(0.077)	(0.027)	(0.030)	(0.028)	(0.029)
Seed	0.115***	0.330	-0.281***	-0.034	0.084*	0.050	0.047
	(0.019)	(0.299)	(0.056)	(0.027)	(0.037)	(0.031)	(0.031)
γ	0.831	ζ	1.496				
γ s.e.	(0.045)	ζ s.e.	(0.247)				
J	16.933						
p-val	0.963						

This table reports GMM estimates aggregating the three labor stages (planting, weeding, and harvesting) into a single labor input. All other moment conditions and estimation details are identical to the baseline specification in Table 1.

Table C8: GMM results without adjustments for unobserved land quality

	α	Const	Inp Intensity	M. Adults	F. Adults	M. Kids	F. Kids
Equip.	0.078***	-0.215	-0.137	-0.056	0.047	0.136	0.182**
	(0.007)	(0.330)	(0.202)	(0.071)	(0.073)	(0.084)	(0.072)
Fert.	0.020**	-0.555	0.194	-0.125	0.071	0.127	0.262**
	(0.007)	(0.574)	(0.152)	(0.081)	(0.086)	(0.111)	(0.108)
Harv. Labor	0.099***	-1.054**	0.052	0.226	0.367	0.034	-0.125
	(0.016)	(0.419)	(0.672)	(0.145)	(0.216)	(0.094)	(0.072)
Land	0.530***	0.636	-1.063***	-0.075	0.056	0.120	0.151**
	(0.039)	(0.362)	(0.194)	(0.062)	(0.064)	(0.071)	(0.065)
Plant. Labor	0.042***	-0.362	-0.442***	-0.033	0.044	0.115	0.152*
	(0.005)	(0.488)	(0.107)	(0.065)	(0.073)	(0.076)	(0.075)
Seed	0.051***	-0.180	-0.417**	-0.049	0.050	0.115	0.133*
	(0.005)	(0.569)	(0.161)	(0.057)	(0.067)	(0.070)	(0.066)
Weed. Labor	0.021***	-0.914*	-0.333***	-0.144	0.117	-0.055	0.364**
	(0.005)	(0.427)	(0.079)	(0.088)	(0.102)	(0.146)	(0.124)
γ	0.842	ζ	2.858				
γ s.e.	(0.048)	ζ s.e.	(0.149)				
J	48.371						
p-val	0.200						

This table reports GMM estimates using raw plot area as the land quantity, without the hedonic adjustment for unobserved land quality described in Appendix E. All other moment conditions and estimation details are identical to the baseline specification in Table 1.

Table C9: GMM results under CRRA utility

	α	Const	Inp Intensity	M. Adults	F. Adults	M. Kids	F. Kids
Equip.	0.207***	-0.785***	-0.257	-0.179	0.312***	0.000	0.019
	(0.019)	(0.194)	(0.173)	(0.099)	(0.089)	(0.070)	(0.054)
Fert.	0.043***	-0.839**	-0.012	-0.090*	0.193***	0.055	-0.060
	(0.009)	(0.262)	(0.051)	(0.045)	(0.053)	(0.043)	(0.034)
Harv. Labor	0.132***	-0.662**	-0.442***	-0.043	0.145*	-0.167***	0.000
	(0.015)	(0.223)	(0.071)	(0.070)	(0.065)	(0.049)	(0.033)
Land	0.264***	-0.452	-0.380**	-0.103	0.263***	-0.014	-0.004
	(0.024)	(0.316)	(0.120)	(0.068)	(0.070)	(0.051)	(0.041)
Plant. Labor	0.073***	-0.948*	-0.391***	-0.124	0.272**	0.014	-0.016
	(0.010)	(0.501)	(0.070)	(0.085)	(0.094)	(0.055)	(0.046)
Seed	0.104***	-0.560	-0.246**	-0.145	0.304**	0.025	-0.000
	(0.012)	(0.452)	(0.102)	(0.084)	(0.095)	(0.061)	(0.048)
Weed. Labor	0.016*	-0.158	-0.188***	0.172***	0.034	0.139	-0.049
	(0.008)	(0.259)	(0.037)	(0.046)	(0.050)	(0.076)	(0.040)
γ	0.839	ζ	1.896				
γ s.e.	(0.045)	ζ s.e.	(0.265)				
J	96.682						
p-val	0.000						

This table reports GMM estimates using CRRA marginal utilities of expenditure rather than CFE demands. All other moment conditions and estimation details are identical to the baseline specification in Table 1.

Table C10: GMM results using all crops

	α	Const	Inp Intensity	M. Adults	F. Adults	M. Kids	F. Kids
Equip.	0.238***	0.816**	-0.231	-0.121*	0.103	0.113	0.163**
	(0.020)	(0.269)	(0.160)	(0.062)	(0.066)	(0.073)	(0.065)
Fert.	0.025**	-0.345	0.407	-0.211*	0.205	0.147	0.307**
	(0.008)	(0.573)	(0.287)	(0.094)	(0.112)	(0.132)	(0.133)
Harv. Labor	0.165***	-0.846**	-0.481***	0.171**	0.210***	-0.061	-0.094*
	(0.021)	(0.263)	(0.080)	(0.059)	(0.060)	(0.058)	(0.042)
Land	0.226***	0.284	-0.628***	-0.096	0.117*	0.074	0.142**
	(0.020)	(0.296)	(0.133)	(0.056)	(0.060)	(0.060)	(0.060)
Plant. Labor	0.062***	-0.051	-0.414***	-0.089	0.129	0.094	0.141*
	(0.007)	(0.483)	(0.091)	(0.063)	(0.074)	(0.071)	(0.072)
Seed	0.114***	0.403	-0.423**	-0.100	0.127	0.110	0.143*
	(0.011)	(0.519)	(0.129)	(0.056)	(0.071)	(0.067)	(0.066)
Weed. Labor	0.023***	-1.205**	-0.318***	-0.206*	0.263**	-0.029	0.364**
	(0.006)	(0.507)	(0.081)	(0.098)	(0.110)	(0.151)	(0.141)
γ	0.853	ζ	2.781				
γ s.e.	(0.042)	ζ s.e.	(0.165)				
J	35.276						
p-val	0.722						

This table reports GMM estimates including all crops rather than restricting to rice plots. All other moment conditions and estimation details are identical to the baseline specification in Table 1.

that observed 0 expenditures can essentially be treated as a missing data problem. While this may appear strong, the assumption essentially requires that welfare can be inferred from observed expenditures and the Frisch elasticities of those goods. See Ligon (2020) and Ligon (2025) for more detail.

What matters for the model in Section 4 is the curvature of utility. The elasticity of λ with respect to total consumption is (minus) the coefficient of relative risk aversion. If this elasticity is constant, then CFE reduces to the nested CRRA case.

E Estimation Details

E.1 Hedonic land quality

I now describe my approach to distinguish true input distortions, unobserved heterogeneity, and noise. Results in both micro and macro literatures recognize the potential for heterogeneous land quality to bias estimation (Benjamin, 1995; Gollin & Udry, 2021). I address this issue using a hedonic approach. Specifically, I train a model to predict rental values from observed plot features on a random sample of rented plots. These features include area, soil type and quality, histories of drought, flood, erosion, and fertilizer application, proximity to water sources, roads, and the household, and (self-reported) sale values.[52] I use cross-validated boosted trees and test the model's fit on a holdout sample, achieving a group-level cross-validated R^2 of 0.22. I then use the model to assign rental values to plots that were cultivated by the owner, for which no rental price is observed. I then use observed and predicted rental prices as a measure of quality-adjusted land quantities.

There are some caveats to this procedure. First, distorted land markets may not accurately reflect true land quality in prices. While this approach allows for land distortions to take the form of either an implicit tax or a ration, it essentially assumes that there is no distortion to the *relative* prices of observable plot attributes, such as soil and proximity to

[52] A similar approach is applied by Gordeev and Singh (2023).

water sources. Nevertheless, there is no a priori reason to assume that relative prices of different attributes should be distorted in a particular direction. Another concern is that transacted plots may be selected on unobservable physical attributes. However, the model would capture the value of these attributes to the extent they are correlated with observable attributes. I also compute results without adjusting for land quality (Table 3, "No Hedonic" column), which remain similar.

E.2 Markov Assumptions for Productivity

In Section 4.1, the Anderson and Hsiao (1981) and Anderson and Hsiao (1982) estimation procedure implicitly assumes that the idiosyncratic component of anticipated productivity ξ_{it} follows a random walk. While this is supported by downstream results in Section 4.2.1, I also compute results assuming a more general Markov process for TFP evolution:

$$\mathbb{E}_{t-1}[a_{it}] = h(a_{it-1}) \tag{38}$$

This means that anticipated (as of t) TFP a_{it} equals households' $t-1$ expectation, $h(a_{t-1})$, plus an orthogonal shock ξ_{it}. In this case, (7) becomes

$$\Delta y_{it} = \sum_k \alpha_k \Delta q_{ikt} + h(a_{it-1}) - a_{it-1} + \xi_{it} + \Delta\phi_{it}. \tag{39}$$

Lagged inputs are still orthogonal to ξ_{it} by the Markov assumption, but the anticipated innovation term $h(a_{it-1}) - a_{it-1}$ must now be proxied for. I use $g(z_{it-1})$, where $z_{it-1} \subseteq \mathcal{I}_{it}$ is a function of households' consumption, investment and other financial variables in households' $t-1$ information sets.[53] This is a valid proxy if households systematically adjust their consumption and investment in anticipation of future productivity shocks[54] and z_{it-1} is rich enough to capture differences in anticipated productivity, which the Townsend data is well

[53] In standard proxy variable approaches in IO (e.g. Gandhi et al., 2020), the assumption of perfect markets allows one to invert the demand function of a flexible input to proxy for the anticipated TFP shock.

[54] This is likely especially relevant when borrowing constraints bind, requiring households to substitute away from consumption and other investments to finance the desired increase in input investment.

suited for. The identification assumption in (8) therefore becomes:

$$\mathbb{E}\left[q_{it-1}\left(\Delta y_{it} - \sum_k \alpha_k \Delta q_{ikt} - g(z_{it-1}) - \Delta\delta_{vt}\right)\right] = 0 \tag{40}$$

with $g(z_{it-1}) = 0$ in the random walk case.

This approach has an important advantage in settings with market failures. Standard proxy-variable methods following Olley and Pakes (1996) assume that a flexible input can be inverted to recover unobserved productivity, which requires TFP to be the only unobservable affecting input demand. This assumption fails when wedges also enter input demands, as they do here.[55] However, in such cases, households' consumption problems become inseparable from their production problem and information about anticipated productivity can be inferred from changes to their consumption and investment patterns.

E.2.1 Implementation

In the main analysis, I assume that $g(z_{it-1}) = 0$, or alternatively households do not anticipate shocks to future TFP (both before or after planting) in prior seasons. In supplementary analysis, I relax this by using a linear function $g(z_{it-1}) \equiv z'_{it-1}\pi$ to proxy for the anticipated productivity shock $h(a_{it-1})$, using variables from households' $t-1$ balance sheets. To avoid overfitting, I apply a multiple post lasso following Belloni et al. (2014) to select only those variables that predict Δy_{it} or any of the Δq_{it}s, the list of which is in Appendix E. This process winds up selecting 14 covariates (Table C6), essentially adding 14 just-identified moments to the estimation. The results are shown in Table C5: Despite none of the estimated $\hat{\pi}$ being individually significant, this produces a J statistic of 54.9 ($p = pval_p i$), suggesting that adding this component worsens the model's ability to match the data. While adding these features to the estimation increases the estimated returns to scale to $\hat{\gamma} = gamma_p i$, this does not lead to substantial changes in the estimates of misallocation in Section 5.

[55] As Shenoy (2021) notes, the market failures that cause misallocation break the "functional dependence" of inputs on anticipated TFP shocks, which would otherwise make proxy-variable methods unidentified.

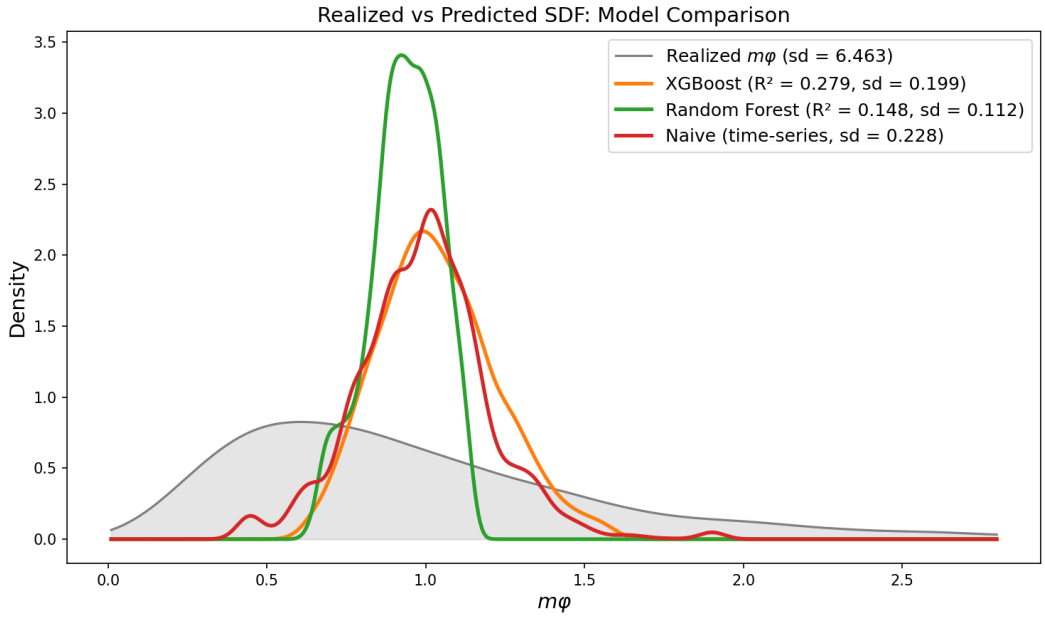
E.3 Machine Learning Predictions

The financial wedge $\hat{\Lambda}$ is defined as the predictable component of the realized Euler residual $\frac{\lambda_{it+1}}{\lambda_{it}}\varphi_{it+1}$. Its variance therefore depends on the specification of the information set projection—that is, on how flexibly one models $\mathbb{E}_t\left[\frac{\lambda_{it+1}}{\lambda_{it}}\varphi_{it+1}\right]$ as a function of observables. As discussed in section 4.1.2, this introduces a bias-variance tradeoff: more flexible models risk overfitting ex-post shocks, while more regularized models compress predictions toward the mean. I assess sensitivity to this choice by comparing two ML projection methods—gradient boosting (XGBoost) and random forest, each tuned via cross-validation and evaluated using 10-fold cross-fitted predictions—alongside the naive time-series benchmark, which uses each household's median realized $\frac{\lambda_{it+1}}{\lambda_{it}}\varphi_{it+1}$ without passing through a prediction step.

Figure E1 shows the distribution of the realized $m\varphi$ alongside the predicted $\hat{\Lambda}$ from each specification. Both ML models produce predictions that are compressed relative to the realized distribution, as expected from the shrinkage inherent in cross-validated prediction. Random forest produces the most concentrated predictions, reflecting its inability to extrapolate beyond training outcomes. XGBoost, which captures nonlinearity through sequential residual-targeting while regularization prevents overfitting, produces a distribution notably similar to the naive benchmark. This suggests that the shrinkage inherent in cross-validated gradient boosting does not substantially attenuate the estimated variance of financial frictions. Despite these differences in prediction variance, the ranking of households by $\hat{\Lambda}$ is broadly consistent across specifications, with pairwise rank correlations of approximately 0.7.

Figure E2 shows that the decomposition of misallocation is qualitatively robust to bypassing the ML prediction step entirely, using instead each household's time-series median as the benchmark for $\hat{\Lambda}$. Financial and input frictions both contribute to misallocation and partially offset each other under this specification as well. The quantitative magnitude of financial misallocation is somewhat larger under the time-series benchmark, as expected: it attributes more of the Euler residual to ex-ante financial frictions since it does not filter out

Figure E1: Realized vs. Predicted $\hat{\Lambda}$ under Alternative Specifications

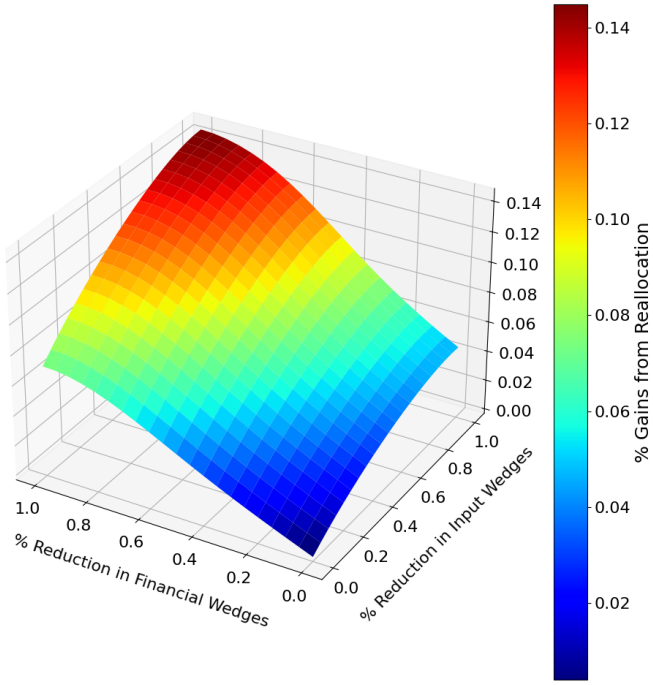


This figure shows kernel density estimates of the realized $\frac{\lambda_{it+1}}{\lambda_{it}}\varphi_{it+1}$ (grey, filled), its cross-fitted predictions under random forest and XGBoost, and the naive time-series median. ML models are tuned via cross-validation and evaluated using 10-fold out-of-fold predictions. Legends report cross-fit R^2 where applicable.

ex-post shocks through cross-validated prediction.

Table E1 provides a direct test of whether the XGBoost predictions are sensitive to the choice of feature set. Row (1) reports the cross-fit R^2 from the baseline specification using 51 features drawn from lagged marginal utilities, household characteristics, and input allocations. Row (2) expands the feature set to 59 variables by adding lagged output and lagged $\hat{\tau}$'s; the R^2 is essentially unchanged (0.282 vs. 0.282), indicating that these additional covariates contain no marginal predictive content for Λ beyond what is already captured. Row (3) provides a stronger test: I take the residual from the baseline prediction and attempt to predict it using the expanded feature set. The resulting R^2 of 0.035 indicates that the baseline residual contains almost no recoverable signal—the baseline model has effectively exhausted the predictable variation in the household information set, at least as proxied by observables. Finally, Row (4) confirms that the estimated productivity shock $\hat{\varphi}$ is similarly unpredictable ($R^2 = 0.015$), consistent with it being a true innovation from the household's

Figure E2: Decomposition of TFP Gains Under Time-Series Benchmark



The figure replicates the decomposition in Figure 5 using the time-series benchmark for $\hat{\Lambda}$ (each household's median realized $\frac{\lambda_{it+1}}{\lambda_{it}}\varphi_{it+1}$), which bypasses the ML prediction step entirely. I compute aggregate TFP under each scenario shrinking Λ and τ towards unity by increments of 0.05. The origin corresponds to the status quo allocation and $(1, 1)$ corresponds to the efficient allocation. The vertical axis shows the percent increase in aggregate TFP relative to the status quo allocation. The figure uses CFE demands and restricts the sample to rice plots, aggregating to the farm level.

Table E1: Residual Unpredictability of ML Predictions

	Target	Features	R^2
Panel A: Financial wedge ($\frac{\lambda_{t+1}}{\lambda_t}\varphi_{t+1}$)			
(1)	$\log\left(\frac{\lambda_{t+1}}{\lambda_t}\varphi_{t+1}\right)$	Baseline (51)	0.282
(2)	$\log\left(\frac{\lambda_{t+1}}{\lambda_t}\varphi_{t+1}\right)$	Kitchen sink (59)	0.282
(3)	$\log\left(\frac{\lambda_{t+1}}{\lambda_t}\varphi_{t+1}\right) - \log(\hat{\Lambda})$	Kitchen sink (59)	0.035
Panel B: Productivity shock (φ)			
(4)	$\log(\psi) - \log(\hat{\xi})$	Kitchen sink (45)	0.015

Notes: Each row reports the cross-validated R^2 from gradient-boosted trees (XGBoost) predicting the indicated target. Baseline features are those used in the main $\hat{\Lambda}$ prediction. Kitchen-sink features add lagged output and lagged $\hat{\tau}$'s. Row (3) tests whether the baseline residual $\log(\frac{\lambda_{t+1}}{\lambda_t}\varphi_{t+1}) - \log(\hat{\Lambda})$ contains predictable signal; $R^2 \approx 0$ indicates the baseline captures the household information set. Row (4) tests whether the productivity shock residual $\log(\varphi) = \log(\psi) - \log(\hat{\xi})$ is predictable; $R^2 \approx 0$ confirms φ is a true innovation. $N = 3424$.

perspective. Together, these results suggest that the main estimates of $\hat{\Lambda}$ are not sensitive to reasonable perturbations of the feature set, and that the cross-fitting procedure successfully prevents the model from attributing ex-post noise to ex-ante financial frictions.